

claude-windows / 9f6357d8-0e35-491a-a830-052e094c7f37

Metadata

```
- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\9f6357d8-0e35-491a-a830-052e094c7f37\subagents\workflows\wf_b6ef6275-de5\agent-a30fb5ee4a473d1cd.jsonl`  
- SHA-256: `bf548e24ff68c36f1640f64477854c83d2de7a370897510a35d76d0132184c52`  
- Source modified: `2026-06-21T13:58:05+00:00`  
- Imported at: `2026-07-05T16:48:25+00:00`  
- project: `wf_b6ef6275-de5`  
- session_id: `9f6357d8-0e35-491a-a830-052e094c7f37`
```

Transcript

1. user (2026-06-21T13:56:13.268Z)

Read the APPLIANCE / CUJ track docs in the khelsutra repo (C:/Users/avidu/Projects/khelsutra) and extract status + next steps. Read: docs/LOCAL_APPLIANCE_ARCHITECTURE.md (the T1?T14 task tracker + \$10 open questions), docs/DRY_RUN_CUJ.md (the DoD ? note Path A + Path B/B-P2 are now DONE, only the droplet box4 remains, which the owner just PARKED), docs/PER_RALLY_SELECTION.md, docs/REQUEST_ACCESS_FORM.md, docs/PORTABILITY.md, and NEXT_STEPS.md. Capture each remaining tracker row / DoD item / next-steps bullet, whether it's actionable or owner-gated, and effort. Note the droplet (option 3) is parked per the owner (GPU machine needs a subscription). Return structured JSON (track="appliance/CUJ").

2. assistant (2026-06-21T13:56:15.834Z)

I'll start by pulling the khelsutra repo to ensure I'm reading the latest state, then read the specified docs.

3. user (2026-06-21T13:56:19.468Z)

```
fatal: Cannot fast-forward to multiple branches.  
---BRANCH---  
master  
---HEAD---  
7750afc docs(dry-run): Path-B live e2e DONE ? in-UI bucket?reel CUJ + screencast (DoD $6 box 3 ?) (#57)
```

4. assistant (2026-06-21T13:56:22.478Z)

The pull failed because the local branch tracks multiple branches. Let me check the state and proceed read-only on the local checkout, which is on master.

5. user (2026-06-21T13:56:26.257Z)

```
---status vs origin/master---  
0      0  
---dirty?---  
?? .claude/  
?? site/access.db
```

6. assistant (2026-06-21T13:56:29.102Z)

Local master is exactly even with origin/master (0/0), clean except untracked files. Good ? the checkout is fresh. Now let me read all the specified docs.

7. user (2026-06-21T13:56:29.975Z)

```

1      # Dry-run CUJ ? UI drives the tool on a bucket video ? reel downloaded local
(multilingual)
2
3      > **Status:** `DRAFT` ? owner-gated . **Owner:** Avi Dullu . **Created:** 2026-06-21 .
**Goal for this session.**
4      > **Umbrella:** a milestone of `docs/LOCAL_APPLIANCE_ARCHITECTURE.md` (the appliance) +
`docs/PER_RALLY_SELECTION.md` (the shipped SPA back-half).
5      > **Honesty note:** grounded in a file:line integration-surface map (this session);
claims tagged `[verified]` / `[gap]`.
6
7      ## 0. The goal (owner's words)
8      Operate the **web UI** ? point at a **video on the GCS bucket** ? the engine (tool)
processes it ? **review/select rallies in the UI** ? **compile** ? **download the reel to the
local machine**. Then **repeat with the UI in Hindi** to prove the multilingual workflow.
9
10     ## 1. What already works vs the gap `[verified]`
11     **? Back half (shipped ? PER_RALLY_SELECTION M0):** the SPA loads `?video=<id>` ? `POST
/api/query` (rallies, default all-selected) ? deterministic query/select ? `POST /api/compile` ?
`<video>` preview + **`<a download>` saves the MP4 locally**. Reel download-to-local **works
today** (`GET /api/highlights/{video_id}/{filename}` streams it; `/api/compile` also returns
`download_url`). Fully **en/kn/hi** (the switcher / `?lang=`).
12
13     **? Front half ? the gap is the SPA, NOT the engine `[corrected 2026-06-21; see §8]:**
14     - The SPA has **no ingest/process UI** ? it can't point at a video (bucket or local),
can't call `/api/process`, can't poll `/api/jobs`. It only gets a `video_id` from `?video=` in
the URL. **This SPA front-half is the *only* net-new work for Path B (= B-P2).**
15     - **HTTP bucket-ingest ALREADY EXISTS in the engine (shipped M2 / [#280]) `[verified
file:line]:** `POST /api/process` accepts a `gs://`/`gcs://` (also `s3://`/`local://`)
`video_path` directly. Submit resolves the URI and **skips the local-existence check for non-
local backends** (`backend/main.py:274,283` ? `storage.resolve_input_ref`; `gs://`/`gcs`
normalized in `backend/storage/ref.py:60`); the job worker then fetches it to scratch
(`_execute_processing` ? `storage.acquire_local_input`, `main.py:159`) and runs the **same**
hash ? validate ? segment ? DB pipeline as a local file ? writing a queryable `video_id` into
the API's SQLite `videos`/`play_segments`. Proven end-to-end by
`tests/test_api_endpoints.py::test_process_nonlocal_uri_skips_exists_and_fetches` (gs:// ? 202,
not 404 ? job `done` ? `db.get_video(video_id)` row with the **source URI preserved**), plus the
hosted-mode-rejection / unknown-scheme-400 / CLI-variant tests.
16     - **Where the earlier "net-new" claim went wrong:** it conflated the **API path**
(`/api/process`, which DID gain URI ingest in M2) with the separate **worker CLI** (`python -m
backend.worker infer --video-uri gcs://?`), which writes `intervals.csv`/`report.json` to the
bucket ? **not** the API DB. The CLI distinction is real; the "the endpoint is local-only"
conclusion was not. ? **Engine-surface claims drift between sessions** (the engine ships PRs
independently) ? re-verify any `[gap]`/`net-new` engine claim against the cited `file:line`
before scoping it as work.
17
18     ## 2. Two paths to the goal
19     **Path A ? Cheapest dry run NOW (~zero new code).** Pre-process the bucket video out-of-
band to mint a `video_id` in the engine DB, then drive the existing SPA back-half. Proves the
**multilingual UI-driven workflow** end-to-end. The only non-UI step is "point at the bucket" (a
one-line CLI pre-process).
20
21     **Path B ? Full "point at a bucket video IN the UI" (SPA-only ? smaller than first
scoped).** The **HTTP bucket-ingest endpoint already exists** (§1) ? so Path B is **only the SPA
front-half** (appliance S1/S2): an **ingest** input (enter a `gs://` URI) ? `POST /api/process`
? an **ingest + progress** screen polling `GET /api/jobs/{id}` ? on `done`, load
`result.video_id` (which the existing back-half already renders). **No net-new engine endpoint**
? B-P1 is already done; the remaining work is **B-P2 (the SPA)**.
22
23     ## 3. Path A runbook (the cheapest dry run) `[verified-steps]`
24     1. **Engine host + deps:** run the engine FastAPI (`python -m backend.main --server`,
binds `127.0.0.1:8000`) on a box with `ffmpeg` + OpenCV. Gemini/motion paths are **CPU-only, no
WSL** ? local Windows works; the CPU droplet works too. (GPU/WSL only needed for native WASB.)
25     2. **Get the bucket video local + into the API DB:**
26     `gsutil cp gs://<bucket>/<match>.mp4 ./incoming/` (or the worker fetch), then `POST
/api/process {"video_path":"<abs local path>","segmenter":"gemini"}` ? poll `GET

```

```

/api/jobs/{job_id}` until `result.video_id`. *(`gemini` needs `GEMINI_API_KEY`; `motion` is key-free but hides fabricated metadata.)*
27     3. **Open the SPA against the engine:** `npm run dev` (Vite proxies `/api` ? `:8000`, same-origin) ? `http://localhost:5173/?video=<video_id>` ? rallies auto-load.
28     4. **Select ? Build ? download:** pick rallies ? Build ? `<video>` preview ? click the download link ? MP4 saved local.
29     5. **Repeat in Hindi:** click `?????` (or `?lang=hi`) ? redo select?compile?download ? confirm the whole flow works with the UI in Hindi.
30
31     **Small unblocked SPA polish that makes Path A more "via the UI" (independent of A/B, useful in both): a ***"Load a video by ID"*** input (paste the `video_id` ? load, instead of editing the URL) + consume the engine's `download_url` (instead of hand-building it). One tight PR.
32
33     ## 4. Open decisions ? OWNER (these gate the dry run)
34     1. **Scope:** Path A (cheapest dry run now) or Path B (the in-UI bucket ingest ? **SPA-only**); the engine endpoint already accepts `gs://`, see §1)? *Recommend A first* (proves the multilingual workflow fast), then B as the appliance build. ? ? **RESOLVED: A first (done, §7), then B (B-P2, the SPA ingest/progress screen).**
35     2. **Compute / detector:** **Gemini** (real rallies; needs your `GEMINI_API_KEY`; uploads downsized chunks to Google = paid + cloud) . **motion** (key-free, runs anywhere, but lower-quality boundaries + fabricated server/shot metadata stays hidden) . **GPU WASB** (best quality, but needs a GPU host + is owner-parity-gated, not production-verified). *Recommend Gemini for "real-ish, cheapest" ? pending your key + OK (paid + uploads footage).*
36     3. **Engine host:** local Windows (simplest, Gemini/motion) . the CPU droplet `168.144.126.176` (shared testbed ? careful) . a GPU VM (only for WASB).
37     4. **Which bucket video:** e.g. `gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4` (already staged, known-good) or another clean ?720p clip (must pass the engine validators).
38
39     ## 5. Honest caveats
40     - **Gemini = paid + uploads footage to Google** (an external/paid action ? needs your explicit OK + a clean clip). **Rotate the chat-shared `GEMINI_API_KEY`.**
41     - **motion** fabricates `server`/`receiver`/`shot_count` (the SPA correctly hides them) ? boundaries are real but metadata isn't.
42     - **GPU/native WASB** is partially wired + owner-byte-parity-gated ? **not production-verified**.
43     - Path A's "point at the bucket" is a **CLI pre-step**, not in-UI ? full in-UI ingest is Path B.
44     - Edge auth: none in the engine (localhost dry-run is fine; required before any non-localhost exposure).
45
46     ## 6. Definition of done (the dry run)
47     - [x] A bucket video ? (engine processes) ? the **SPA loads its rallies** ? select ? compile ? **reel downloaded to the local machine**, all driven from the UI (Path A allows a one-line CLI pre-process). ? ? **Path A done (local, 2026-06-21)**, see §7.
48     - [x] The **same flow works with the UI in Hindi** ? multilingual workflow confirmed. ? ? (Kannada also works via the same switcher.)
49     - [x] *(Path B = B-P2, SPA-only)* the bucket video is pointed-at **from the UI** (no CLI pre-step). ? ? **DONE (live e2e, 2026-06-21), see §9.** In the SPA: entered a `gs://` URI ? `POST /api/process` ? polled `/api/jobs` ? **12 rallies loaded** ? Build ? **reel downloaded (1080p, 76s)**, all from the UI. Surfaced + fixed an engine compile bug along the way (engine [#289](https://github.com/Khelsutra/badminton-highlight-indexer/pull/289)). **Screenshot:** `gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screenshot.mp4`.
50     - [ ] *(option 3)* the same CUJ confirmed on the **CPU droplet** (Linux deploy target) ? droplet reachable but currently **un-provisioned** (no engine checkout; ~3.8GB RAM) ? needs a from-scratch engine install or a proper deploy of the fixed engine.
51
52     ## 7. ? Run result ? Path A, local (2026-06-21)
53     Ran the real CUJ end to end on the local Windows box, isolated engine off
    `origin/master` (fresh DB/output worktree), **Gemini** detector:
54     1. `gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4` (1080p/30fps) ?
    `gsutil cp` local ? `POST /api/process {segmenter: gemini}`.
55     2. Engine processed via Gemini (3 chunks) ? **12 rallies** (16 found, 4 too-short rejected), `video_id 05e0e577`.

```

56 3. SPA (`npm run dev`, Vite proxies `/api`?:8000) opened at `?video=05e0e577` ?
 loaded the 12 real rallies ("Loaded 12 rallies from this session").

57 4. **Build ? `/api/compile` ? reel** (1920x1080, 76.5s) ? **downloaded to
 `~/Downloads/khelsutra-dryrun/highlights\_local\_en.mp4`** (50.9 MB).

58 5. **Repeated with the UI in Hindi** (`?lang=hi`): the page + rally cards + "??? ?????"
 + "???? ???? ???? ??" all in Hindi; reel compiled + **downloaded** (`highlights\_local\_hi.mp4`).
 ? multilingual workflow confirmed.

59

60 Screencast/proof: per-locale screenshots; both reels on the local machine. Screencast of
 the i18n SPA is at `gs://khelsutra-rally-corpus/promo/khelsutra-i18n-screencast.mp4`.

61

62 ### ? Fix the dry run surfaced (engine) ? MERGED-PENDING

63 The async-job server **hung every job in `queued`** when run the documented way (`python
 -m backend.main`): main.py runs as `__main__` (sets `__main__.db_instance`), but `job_worker`
 reads `import backend.main` ? a **separate** module whose `db\_instance` is `None` ? the drain
 crashes silently. Latent because the unit tests monkeypatch the executor inline and the UI was
 verified vs a stub engine. **Fix + a real-subprocess regression test ? engine
 [#281](https://github.com/Khelsutra/badminton-highlight-indexer/pull/281)** (dispatch through
 `backend.main`; FAILS-without / PASSES-with). Unblocked the run via the single-module
 invocation; the fix applies to the droplet too.

64

65 ## 8. ? Correction (2026-06-21): B-P1 (the bucket-ingest endpoint) was already shipped ?
 verified against code

66 **What this doc claimed (\$1, original):** "`POST /api/process` accepts only a **local**
 `video\_path`" and "bridging `gs://` ? a queryable `video\_id` in the API DB is **net-new**
 (T12/T14)."

67

68 **What the code actually does `[verified file:line, this session]`:** `POST
 /api/process` has accepted bucket/Drive URIs since **engine M2
 ([#280](https://github.com/Khelsutra/badminton-highlight-indexer/pull/280))**. A
 `gs://`/`gcs://`/`s3://`/`local://` `video\_path`:

69 1. is resolved at submit (`backend/main.py:274` ? `storage.resolve\_input\_ref`;
 `gs://`/`gcs` at `backend/storage/ref.py:60`; explicit scheme wins under default
 `input\_backend=local` at `ref.py:87`),

70 2. **skips the local-`os.path.exists` gate** for non-local backends (`main.py:283`) ?
 returns **202, not 404**,

71 3. is fetched to scratch by the worker (`\_execute\_processing` ?
 `storage.acquire\_local\_input`, `main.py:159` ? `GCSStorage.fetch\_to\_local`), then

72 4. runs the **same** pipeline as a local file, writing a queryable `video\_id` to the API
 SQLite DB.

73

74 **? Scope pin (default single-user mode):** this holds with
 `security.allowed\_video\_root` **unset** (the local-appliance default). In **HOSTED mode**
 (`allowed\_video\_root` set), a bucket URI is **correctly rejected `400`** ? a tenant can't point
 the engine at an arbitrary bucket. That is by design, not a regression.

75

76 **Already tested** (so this can't silently regress):
 `tests/test\_api\_endpoints.py::test\_process\_nonlocal\_uri\_skips\_exists\_and\_fetches`,
 `?:test\_process\_hosted\_mode\_rejects\_nonlocal\_uri` (the hosted-mode counterweight),
 `?:test\_process\_unknown\_scheme\_returns\_400`,
 `?:test\_cli\_nonlocal\_input\_fetches\_and\_keeps\_uri`. So **B-P1 needs zero new engine feature
 code** ? the only Path-B work is **B-P2 (the SPA ingest/progress screen)**. *(The hermetic test
 fakes the network download via `m.storage.fetch`, so it proves the CODE routing, not the live
 GCS fetch ? that needs `google-cloud-storage` + ADC on the box, verified at deploy time / option
 3.)*

77

78 **Why it went stale (anti-spiral lesson):** the \$1 claim was tagged `[verified]` but was
 inferred from the **worker CLI** (`backend.worker infer`, which writes CSV to the bucket) rather
 than checked against the **API endpoint's** routing code ? and M2/#280 had since wired URI
 ingest into `/api/process`. **The engine ships PRs between sessions, so its surface drifts.**
 Convention going forward (also baked into the memory steering): **before scoping any
 `[gap]`/`net-new` engine claim as work, re-verify it against engine code at a cited `file:line`
 ? a `[verified]` tag without a `file:line` is a hypothesis, not a fact.**

79

80 ## 9. ? Run result ? Path B, live in-UI bucket?reel CUJ (2026-06-21)

```

81     The full Path-B flow, driven entirely from the UI81 (no CLI pre-step), against a real
local engine (`python -m backend.main --server`, Gemini detector,
`GOOGLE_APPLICATION_CREDENTIALS` for the GCS fetch):
82     1. Opened the SPA ? the "Process a video" ingest panel (B-P2) ? pasted
`gs://khelsutra-rally-corpus/videos/mahadevpura_smoke_180s.mp4` ? "Process video".
83     2. The engine fetches the bucket video83 (`google-cloud-storage` + ADC) ? "Gemini"
segmented it (live "Processing? segmenting (gemini)" progress in the UI) ? "12 rallies"
(`video_id 05e0e577?`), handed off via `video=` ? the existing back-half rendered them.
84     3. "Build reel" ? `/api/compile` ? "reel preview + download" ? MP4 saved locally
("1920x1080, 76s").
85     4. "Screenshot of the whole CUJ" (Playwright-recorded; the ~3-min Gemini wait fast-
forwarded): `gs://khelsutra-rally-corpus/promo/khelsutra-bucket-cuj-screenshot.mp4`.
86
87     ### ? Bug the run surfaced + fixed (engine) ? MERGED
88     `/api/compile` "500'd" on a bucket-ingested video: `videos.filepath` keeps the
"source URI" (`gs://`, M2 provenance), and the stitcher was handed that directly ? ffmpeg
can't open a `gs://` path. "Fix:" compile now resolves the source via
`storage.acquire_local_input` first (identity for a local path; fetch-to-scratch for a bucket
URI) + a regression test ? engine "#289"(https://github.com/Khelsutra/badminton-highlight-indexer/pull/289)89 (full suite 1182?). This is what makes the in-UI bucket?reel CUJ actually
complete. "Proof-and-validation-driven: recording the screenshot is what found the bug."
89
90     ### Changelog
91     - `2026-06-21` ? created; Path-A run recorded ($7).
92     - `2026-06-21` ? "$8 correction:" verified B-P1 (`gs://`?`/api/process`?API-DB
`video_id`) is already shipped + tested (M2/#280); $1/$2/$4/$6 reframed so Path B = SPA-only
(B-P2). Sibling stale claims fixed in `LOCAL_APPLIANCE_ARCHITECTURE.md` (`/api/health` exists;
bucket-URI ingest not a net-new endpoint; T12 engine-half done) + `README.md`.
93     - `2026-06-21` ? "$9: Path-B live e2e DONE" ? the in-UI `gs://`?reel CUJ ran end to
end (12 rallies ? reel downloaded); surfaced + fixed the compile-from-bucket 500 (engine #289);
screenshot uploaded to the promo bucket. "DoD $6 box 3 ?."
94

```

8. user (2026-06-21T13:56:30.098Z)

```

1     # Single-box appliance ? UI-powered rally indexer + reel generator (KhelSutra Studio)
2
3     > "Status:" `DRAFT` ? owner-gated (auth choice, retention, cost ceiling, GPU host, P5
bridge) . "Owner:" Avi . "Created:" 2026-06-20 . "Last updated:" 2026-06-20
4     > "Lifecycle:" `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5     > "Tracking anchors:" $9 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
6     > "Relation:" peer-of `PER_RALLY_SELECTION.md` / `REQUEST_ACCESS_FORM.md` in this
repo; "consumes" the engine's decoupled `StorageBackend` + `python -m backend.worker` seams
and the FastAPI loop; "extends" the engine's `HOSTING_PLAN.md` gating + `UI_PROPOSAL.md` alpha
screens. Engine-side changes here are "proposals to coordinate" with the active
decoupled/worker sessions ? not written unilaterally.
7     > "Honesty note:" every engine-fact claim is tagged `[verified]` (read against engine
source this session), `[design]` (proposed here), or `[researched]` (needs sign-off). Not
legal/financial advice.
8
9     ---
10
11     ## 0. TL;DR
12     "KhelSutra Studio" turns the engine from a CLI into a "single-box, UI-powered rally
indexer + reel generator": the khelsutra `web/` React SPA, served behind an "edge auth gate",
drives the "existing" engine FastAPI loop (ingest ? index ? pick rallies ? reel ? download) on
"one machine", with `StorageBackend=local` for the MVP. It is the "single-box deployment
profile of the just-shipped decoupled architecture" ? storage=local + compute=local is the
degenerate collapse of the same storage/worker split.
13
14     The "single most important caveat:" the engine has "zero app-layer auth and CORS"
95 "[verified backend/main.py:42]", so safety is structural "only" because the engine stays
bound to `127.0.0.1:8000` 96 "[verified backend/main.py:638]", a firewall blocks inbound `8000`,

```

and a reverse proxy/tunnel is the sole authenticated ingress ? **the gate, not the app, is the tenancy boundary.**

15

16 Two honest hardware truths: (1) the current droplet is **CPU-only** ? real detection there is **Gemini (cloud)** or **stub (CPU mock)**; **real local-GPU detection** needs a GPU host (native WASB, in-flight). (2) The owner priority is to **stand up a GPU host ASAP** ? provider-agnostic (DO GPU droplet / GCE GPU VM / Cloud Run+GPU) ? but that runs **in parallel** with the UI MVP, which ships on Gemini today.

17

18 **## 1. Problem & goal**

19 The decoupled engine just shipped a ``StorageBackend` ABC + a `python -m backend.worker {infer|train}` dispatcher `[verified backend/storage/, backend/worker/__main__.py], and the FastAPI loop already covers `process ? jobs ? query ? compile ? highlights` `[verified backend/main.py]. What's missing is a console that lets a non-CLI operator run the whole loop on a single box, honestly, behind a gate.`

20

21 **Headline CUJ (owner's words):** a **hosted website** where I pick an input ? **a Google Drive file, a bucket file (``s3://`/`gs://``), or a local upload** ? it lands in the **bucket (DO Spaces / GCS)** or locally ? I **generate highlights** on a **GPU-enabled host** running the real e2e ? I **pick rallies, watch, and download** ? **completely from the UI**.

22

23 **Outcome ("good"):** an operator signs in, picks a source, watches it index, curates rallies, downloads a reel ? on the **live CPU droplet today** (Gemini/stub), and on a **GPU host later** (native WASB) with **no UI fork**.

24

25 **Read to get current:** ``PER_RALLY_SELECTION.md`` (§3/§4/§10 contract anchors), engine ``HOSTING_PLAN.md`` (gating), ``DECOUPLED_COMPUTE/HANDOFF.md`` (what's built), ``VIDEO_LOCALITY_MODEL.md`` (privacy/retention).

26

27 **## 2. Decisions locked**

28 | # | Decision | Source / date | Implication |

29 |---|---|---|---|

30 | D1 | Reverse-proxied **single origin**; engine stays ``127.0.0.1:8000`` | ``[verified main.py:638]`; 2026-06-20 | CORS ``*`` is moot from the browser ? the proxy is the boundary |

31 | D2 | **Edge auth**, zero engine auth code | ``HOSTING_PLAN.md``; 2026-06-20 | Caddy basicauth OR cloudflared + Cloudflare Access |

32 | D3 | MVP drives the **in-process FastAPI worker** loop, ``storage=local`` | ``[verified job_worker.py:37, local.py:33]`; 2026-06-20 | bucket worker = reserved scale path, out of MVP |

33 | D4 | Embed **RallyPicker** from ``PER_RALLY_SELECTION.md`` **verbatim** | PR [#11]; 2026-06-20 | the console is a superset; do not redesign the picker |

34 | D5 | Console lives in the khelsutra ``web/`` React scaffold | ``web/README.md``; 2026-06-20 | Cloudflare-Pages-style root, not engine static |

35 | D6 | Compute placement is a **setting**, rendered from ``GET /api/capabilities`` | ``[design]`; 2026-06-20 | UI offers ONLY what the box can honestly do |

36 | D7 | **Ingest** is a source menu (local upload · bucket URI · Drive) normalizing to one video object | ``[verified backend/storage/ref.py parse_uri]`; 2026-06-20 | Drive is a **stub** today ? scoped honestly (§4.2) |

37 | D8 | GPU host is **provider-agnostic** (DO droplet / GCE VM / Cloud Run+GPU); storage stays the bucket | Owner 2026-06-20 | compute and storage are separate planes; no provider lock-in |

38 | D9 | **Observability** is first-class ? structured logs on every new path + a DoD item | Owner 2026-06-20 | see §5; verified during CUJ replay |

39 | D10 | Delivery = a **self-hosted webserver** + UI the user spins up (NOT a desktop app); the user points at **local storage + GPU/CPU OR cloud locations** and the UI drives the tool | Owner 2026-06-20 | **supersedes** ``DESKTOP_APP_PLAN.md``; one artifact (this appliance) runs on the owner box, a droplet, or a GPU host ? same UI, compute/storage are settings |

40

41 **## 3. Foundation ? what the engine gives us today** ``[verified this session unless noted]`

42 - **FastAPI loop:** ``POST /api/process`` (202 + ``job_id``) ? poll ``GET /api/jobs/{id}`` ? ``POST /api/query`` ? ``POST /api/compile`` ? ``GET /api/highlights/{video_id}/{filename}`` (FileResponse) ``[main.py]`. **Chunked upload exists** ? ``POST /api/upload/init`` · ``PUT /api/upload/{id}/part/{i}`` · ``POST /api/upload/{id}/complete`` · ``DELETE /api/upload/{id}`` ``[verified uploads.py:64,89,123,147]` ? but the bundled ``app.js`` **never calls them** (it only hits ``/api/process`` + ``/api/compile`` ``[verified app.js:88,373]`). So browser upload is **client-**

```

wiring only**, no engine change.
43     - **In-process worker:** one module-global `ThreadPoolExecutor(max_workers=1)`
  `[verified job_worker.py:37]` ? strictly **one job at a time**; the DB `jobs` table is the
  clock; `fail_stale_jobs` sweeps queued/running on boot. Gemini chunk concurrency is a
  **separate** knob `indexing.ai_max_workers` (**default 5** `[verified models.py:183]`; set to 1
  to serialize) ? it is **not** already 1.
44     - **Storage:** `storage.backend` default = `"local"` `[verified models.py:298]`;
  `LocalStorage.fetch_to_local` is **identity** `[verified local.py:33]`, `put` no-ops on
  `src==dst` ? so on a single box the worker's pull?pipeline?push **collapses to read-local/write-
  local**. URI scheme `parse_uri` `[verified ref.py]`: bare/`local://` ? local; `s3://`
  (AWS/R2/**DO Spaces**/MinIO, by endpoint) ; `gcs://`/`gs://` ; `gdrive://` ? **follow-up stub
  (emulated only)**. Credentials never in config ? boto3 default chain / `~/aws` `[verified
  s3.py]`.
45     - **Compute seam:** `indexing.detector_impl` ? `{auto`=WSL, `native`=Linux-native,
  `stub`=CPU mock}; default `"auto"` `[verified models.py:129]`, resolved in
  `trajectory_hybrid._resolve_runner` `[verified :76-85]`. **Native WASB is in-flight on branch
  `feat/native-wasb-runner`:** `native_wasb_runner.py` **exists** `[verified]`;
  **`fusion_hybrid._build_native_runner` returns `NativeWasbRunner`** `[verified
  fusion_hybrid.py:86]` while **`trajectory_hybrid._build_native_runner` still raises** `[verified
  :57-63]`. Acceptance gate = **owner byte-parity** (WSL CSV == native CSV) on a real GPU box ?
  not exercisable on the droplet or in the offline suite. Treat native as **not production-
  verified**.
46     - **Worker CLI:** `python -m backend.worker {infer|train}` pulls a `--video-uri` ?
  pipeline ? pushes `outputs/<video_id>/{intervals.csv,report.json}` `[verified
  worker/___main__.py:88,149]`. The validated GPU-free/key-free combo: `detector_impl=stub` +
  `--set indexing.skip_ai_handoff=true` + `storage.s3`.
47     - **Gating:** cloudflared Tunnel (outbound-only, no inbound ports) + Cloudflare Access
  email-OTP; backend trusts `Cf-Access-Authenticated-User-Email` `[HOSTING_PLAN.md]`. `GET
  /api/health` **now EXISTS** (engine M1, [#268]) ? `{status, sport, default_segmenter}`
  `[verified main.py:127, corrected 2026-06-21 ? was "does not exist"]`.
48     - **Droplet reality:** `168.144.126.176`, **CPU-only, NO GPU**, region `blr1`; co-hosts
  the marketing site (`/srv/khelsutra-site`, node `:8787`) **and** the engine (`~/rally`); ran the
  full train+infer e2e against DO Spaces with the **GPU mocked** `[SETUP_NEW_MACHINE.md,
  DECOUPLED_COMPUTE/HANDOFF.md]`. Shared testbed: MinIO `9000/9001` + GCS emulator `9023` also run
  there.
49     - **Absent (net-new):** `GET /api/capabilities`, `GET /api/videos` (only single-row
  `get_video(id)`, `GET /api/reels/{video_id}`, any source/proxy streaming endpoint, any
  **Dockerfile**, any per-video **cost ledger** `[verified ? grep returns none]`.
50
51     ## 4. Design / architecture
52
53     ### 4.1 Topology & request path
54     ONE box: **edge gate** (cloudflared+Access or Caddy basicauth) ? **reverse proxy /
  static host** serving the `web/` SPA bundle and proxying `/api/*` ? engine **FastAPI
  `127.0.0.1:8000`** ? **LocalStorage** ? **detector seam** (Gemini/stub today; native WASB on a
  GPU host). The firewall blocks inbound `8000` + the test ports `9000/9001/9023`. The marketing
  site stays a **separate** systemd unit on a **separate** subdomain.
55
56     ...
57     PUBLIC INTERNET
58         ? (1) EDGE GATE ? the ONLY ingress; auth lives HERE, not in the app
59         ? cloudflared Tunnel + Cloudflare Access (no inbound ports) OR Caddy :443 +
  basicauth
60         ?
61         ?????????????????????????????????????????????????????????????????????????????????????
62         ? ONE BOX   firewall BLOCKS inbound 8000 + 9000/9001/9023                                     ?
63         ? (2) reverse proxy / static host (Caddy/cloudflared)                                     ?
64         ?         /, /assets/* ? web/ React SPA         /api/*, /api/highlights/* ? :8000 ?
65         ?         ? ONE ORIGIN to the browser ? engine CORS '*' is irrelevant ?
66         ?         ?                                     ?
67         ? (3) ENGINE FastAPI (host=127.0.0.1:8000, HARD-bound [main.py:638]) ?
68         ?         control: jobs table + ThreadPoolExecutor(max_workers=1) drain ?
69         ?         data: uploads.py chunked upload · /api/compile · /api/highlights ?
70         ?         ?                                     ?
71         ? (4) STORAGE = LocalStorage (default) fetch=identity · put=no-op(src==dst) ?

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72      ?           ?
73      ? (5) COMPUTE seam: gemini (cloud, key) · stub (CPU mock) [droplet]
74      ?           native WASB (GPU host) · auto (owner Win+WSL)
75      ?????????????????????????????????????????????????????????????
76      RESERVED SCALE PATH (out of MVP): console "compute target" selector ? storage=s3 (DO
Spaces)
77      ? SEPARATE GPU host runs `python -m backend.worker infer --video-uri s3:///?` ?
outputs/<vid>/ back
78      (needs a NET-NEW orchestration endpoint reflecting outputs into the jobs/poll
contract)
79      ...
80
81      ### 4.2 Ingest sources (the headline) ? one menu, one normalized video object
82      All sources normalize to ***a video the pipeline can read*** ? a local path (MVP) or a
`StorageRef` (scale):
83
84      | Source | Path today | Status |
85      |---|---|---|
86      | **Local upload** | SPA chunks the file ? `max_part_mb` via the existing
`/api/upload/init\part\complete` `[verified uploads.py]` ? server path ? `POST /api/process`.
(Scale: also `StorageBackend.put` to `s3://`/`gcs://`.) | **wire client only** (endpoints exist)
|
87      | **Bucket URI** (`s3://`, `gs://`) | **POST /api/process` accepts the bucket URI
directly** (resolve ? fetch to scratch ? same pipeline ? API DB) `[verified main.py:159,274,283
? corrected 2026-06-21; was "net-new"]`; the worker CLI also fetches `--video-uri`. So the
**engine endpoint is NOT net-new** ? only the SPA call that POSTs the URI + polls `/api/jobs`
is. *(Default single-user mode; HOSTED mode w/ `allowed_video_root` correctly rejects a bucket
URI 400.)* | **engine: DONE (verified); SPA ingest call: design (= B-P2)** |
88      | **Google Drive** (`gdrive://`) | `parse_uri` recognizes it, but the gdrive backend is
an **emulated stub** `[verified ref.py, DECOUPLED_COMPUTE research]` | **stub ? roadmap; MVP
requires a Drive?bucket pre-step; do **not** present Drive as live until the backend is real |
89
90      ### 4.3 Compute placement matrix (one UI, surfaced from `/api/capabilities`)
91      | Placement | Runs | Needs | Real/mock | Honest caveat |
92      |---|---|---|---|---|
93      | **CPU droplet ? gemini** (MVP target) | `segmenter=gemini`, in-proc worker |
`GEMINI_API_KEY`; network; **no GPU** | **REAL** | Cloud brain: uploads ?1280px chunks to Google
? **must be in consent copy**; **no per-video cost ceiling** today |
94      | **CPU droplet ? stub** | trajectory hybrid + `detector_impl=stub`+`skip_ai_handoff` |
nothing | **MOCK** | not real tracking ? UI must label "stub (CPU mock ? not real detection)";
demo/regression only |
95      | **CPU droplet ? motion** | `segmenter=motion` | nothing | REAL, low-quality | only no-
key/no-GPU real segmenter; never surface `server/receiver/events` (fabricated) |
96      | **GPU host ? native WASB** (in-flight) | `wasb`+`detector_impl=native` ?
`NativeWasbRunner` (in-proc torch) | a **GPU host ? DO GPU droplet / GCE GPU VM / Cloud
Run+GPU**; `WASB_WEIGHTS` | **REAL**, on-device | partially wired (fusion builds it, trajectory
refuses); **owner byte-parity gate**; **Cloud Run/GCE need a containerized worker ? no
Dockerfile exists, and WASB's torch-1.11 conda env makes the image non-trivial** |
97      | **Owner Win+WSL ? auto** | `wasb`+`detector_impl=auto` (WSL2+conda) | Windows+WSL2+GPU
| REAL (today) | Windows-only transport; the working real-GPU path for owner verification |
98      | **Remote GPU worker via bucket** (reserved) | console CPU + `storage=s3`; separate GPU
host runs `backend.worker infer` | bucket creds; a GPU host; a **net-new orchestration
endpoint** | REAL (remote) | FastAPI loop and worker CLI are **separate** today; "remote GPU as
a setting" is **design** ? selector shown **disabled** in MVP |
99
100     **Provider note:** **Cloud Run + GPU** (scale-to-zero, pay-per-job) is the strongest fit
for the engine's own **per-video-billing** economics (`DECOUPLED_COMPUTE` #246) ? gated on the
**containerization feasibility check** above.
101
102     ### 4.4 Auth / exposure (non-negotiable invariant)
103     No engine auth, by design; the gate is external. **Invariant (all three must hold):**
engine bound to `127.0.0.1` + firewall closes `8000` + edge gate up. Go-live checklist
`[HOSTING_PLAN.md:102-108]`: Access/basicauth ON; **rotate the chat-shared `GEMINI_API_KEY`**;
set `security.allowed_video_root`; rate-limit; per-user quota.
104

```



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105     ### 4.5 UI operator console (web/ SPA ? superset of RallyPicker)
106     - **S1 Ingest & Process** ? the source menu (§4.2) + a segmenter picker from
`/api/capabilities` ? `POST /api/process`.
107     - **S2 Progress** ? 3s poll of the free-text stage
(`hashing/validating/segmenting/finalizing`), a one job at a time notice, and the restart-
failure terminal state.
108     - **S3 Rally pick ? reel** ? embed RallyPicker verbatim (`PER_RALLY_SELECTION.md`):
honor `min_shot_count`; never render motion `server/receiver/events` or a fake confidence meter.
Per-rally is ONE screen of the console.
109     - **S4 Reels / History** ? needs `GET /api/videos` + `GET /api/reels/{video_id}` for
reload-survival.
110     - **S5 Settings** ? render `config` + `capabilities`; the compute/storage target
selector shown DISABLED ("single-box: local") until the bucket-worker bridge lands.
111
112     ### 4.6 Reuse map
113     Reused (no re-plumb): the full `/api` loop; the in-process worker; the chunked
upload endpoints (un-called); LocalStorage default; the detector seam incl. the landing
`NativeWasbRunner`; the RallyPicker design + anchors; the `web/` scaffold; the
`HOSTING_PLAN` gate; the `SETUP_NEW_MACHINE` runbook. Net-new: Caddyfile + systemd units;
`GET /api/capabilities`; `GET /api/videos`; `GET /api/reels/{video_id}`; `GET /api/health`; the
SPA console screens; ingest-from-bucket wiring; a retention sweep; (reserved) the bucket-worker
bridge.
114
115     ## 5. Observability ? logging & monitoring `[required, first-class]`
116     Launch-readiness directive (owner 2026-06-20): logs and monitoring taken seriously,
with a full log-analysis pass across every new code path. The engine already uses a
centralized Python logger `[verified main.py:12-17]` ? carry that discipline into every new
path.
117
118     - Correlation ID end-to-end. Thread the existing `job_id` (from `/api/process`)
plus an `upload_id`/`request_id` through ingest ? process ? jobs ? compile ? highlights, and
into the worker for the bucket path, so one CUJ is traceable across UI ? FastAPI ? worker ?
GPU host. Log it on every line.
119     - Structured logs. New endpoints/screens emit structured (JSON-able) logs at entry
/ result / error ? not just "it ran". The SPA logs client-side errors + failed fetches with
the same correlation id.
120     - Loud failures, never silent drops. Carry the engine's ethos (#240 "surface failed
chunks loudly") + RallyPicker's `min_shot_count`-warning rule: a partial/empty reel must
never look complete; every dropped clip/chunk is logged + surfaced in the UI.
121     - Health + metrics. Ship the missing `GET /api/health` `[HOSTING_PLAN.md:42]` and a
thin metrics surface (jobs queued/running/failed, last error, disk free, per-video cost once a
ledger exists) for the engine-mode banner + alerting.
122     - Aggregation for transient hosts. A scale-to-zero Cloud Run / GPU host is
ephemeral ? logs must ship to a durable sink (the bucket or a log service), not die with the
instance.
123     - Log-analysis pass before merge (DoD item). Every PR that adds a code path includes
a check that the path logs meaningfully, and the logs are inspected during CUJ live-replay
(§8 Layers 173). Ties [[feedback-launch-quality-bar]].
124
125     ## 6. Risk table
126     | Risk | Today | Mitigation |
127     |---|---|---|
128     | No app auth + CORS `` `[main.py:42]` | open if any of {localhost-bind, firewall,
gate} slips | enforce all three; LIVE posture test that raw `:8000` is refused from outside
|
129     | Test ports on shared droplet | MinIO `9000/9001` + GCS emu `9023` | firewall closes
them |
130     | Retention | `output/+`uploads/` originals never deleted | sweep / keep-proxy-only
before any non-owner exposure |
131     | Surprise Gemini/GPU bill | no cost ceiling/ledger | per-video cap;
`ai_max_workers` (default 5) tunable; `max_workers=1` job serialization |
132     | Native WASB unverified | partially wired; owner-parity-gated | treat as not-
production-verified until owner byte-parity on a real GPU box |
133     | New engine endpoints | `capabilities`/`videos`/`reels`/`health` don't exist |
coordinate engine PRs with active sessions; branch from `origin/master`, re-verify HEAD,

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explicit `git add`** |
134 | Drive ingest overstated | `gdrive://` is a stub | don't present Drive as live; require
Drive?bucket pre-step until real |
135 | No Dockerfile | runbook-only reproducibility | feasibility-check a worker image before
promising Cloud Run/GCE |
136 | Shared-checkout collisions | active worker sessions share the engine tree | the
branch-safety protocol on every engine-side PR |
137 | **Provisioning-credential leak** | a DO API token / cloud creds could leak if pasted
in chat or committed (cf. the **already-rotated Spaces key** exposed in chat,
`DECOUPLED_COMPUTE/HANDOFF.md`) | authenticate the CLI **on-box** (`doctl auth init` interactive
/ `gcloud` already authed) so the secret never enters chat or the repo; least-privilege token,
revoke after; creds 0600, never committed; **creating paid GPU infra is a confirm-first action
every time** |
138
139 ## 7. Honest limits ? what this does NOT do
140 Does **not** add multi-tenant identity/RLS (single-tenant alpha; uploads/outputs
global). Does **not** run real WASB on the CPU droplet (Gemini or stub only). Does **not** make
the remote-GPU bucket path work (design; selector disabled). Does **not** provide in-UI source
playback, job cancel, or a true progress bar (coarse free-text stage, poll-only). Does **not**
containerize (runbook; no Dockerfile `[verified]`). Does **not** verify native WASB (owner byte-
parity on a real GPU box, not this offline suite). A green SPA + contract tests prove the
**console plumbing**, not engine rally-detection quality (a separate track).
141
142 ## 8. CUJs & live-test plan `[required]`
143 **CUJs:** **CUJ-1** ingest (Drive/bucket/local) ? index ? reel ? download (core);
**CUJ-2** re-open a past match after reload; **CUJ-3** pick-by-query (deterministic filters +
honesty-hiding); **CUJ-4** settings/compute placement (truthful capabilities).
144
145 **Live-test plan ? 5 layers, every CUJ replayed LIVE throughout development:**
146 - **L1 ? Mock engine (hermetic, fast, default + CI).** A tiny FastAPI/MSW implementing
the **exact** contract (upload; `process` 202; `jobs` state machine incl. restart-failure;
`query` with a motion fixture for honesty-hiding; `config`; `capabilities` per box profile ?
CPU-no-key / CPU-with-key / GPU; `compile`; `highlights` tiny mp4; the new `videos`/`reels`).
The SPA drives **all 4 CUJs in <1s**, **replayed via the Claude Preview/browser tools** (start
dev server ? fill the ingest picker ? click through process/poll/RallyPicker ? assert the reel
link via snapshot/network). Catches reload-survival (CUJ-2), the `play_segments` field-name bug
(CUJ-3), capability honesty (CUJ-4).
147 - **L2 ? Engine-contract checks (real FastAPI, GPU-free).** Boot `python -m backend.main
--server` with `storage=local` + `detector_impl=stub` + `skip_ai_handoff` and assert the **mock
and the real engine agree shape-for-shape** (202 keys, jobs enum, `result.play_segments` schema,
compile `{status,filepath}`, highlights content-type). Every new endpoint is born with a mock
handler **and** a real-engine contract test **in the same PR**. Runs in CI.
148 - **L3 ? Real local e2e (the MVP substrate).** Through the gate on the droplet: CUJ-1
with `stub` (key-free) and `gemini` (real, short clip to bound cost); the **exposure-safety
posture test** (raw `:8000/`:900x` refused from outside) as an assertion, not an assumption;
reuse the `parity.yml` dual-host pattern.
149 - **L4 ? CI regression (green gates merge).** `npm test` + a stub-engine contract job +
headless browser-replay CUJ scripts; one small PR per tracker row off `origin/master`, **no
stacking**; coverage at/above the floor; **logs inspected during replay** (§5).
150 - **L5 ? Owner hardware e2e (handed off).** Native WASB **byte-parity** on a real GPU
box; real-4K **golden-corpus** runs (Gemini + native); Modern-Standby/`cv2`-vs-`ffmpeg`
behavior; a multi-GB browser upload. Reported back into §9, stated plainly as **not verified**
in the offline suite.
151
152 ## 9. Deliverables & progress tracker ? **source of truth**
153 Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking; CI green gates merge; coverage ? floor; each PR
ships its tests + meaningful logs (§5).
154
155 | ID | Deliverable | Depends on | Gated? | Status | PR |
156 |----|-----|-----|-----|-----|----|
157 | T0 | This design doc + `docs/README.md` + `NEXT_STEPS.md` pointers | ? | owner | ? |
(this PR) |
158 | T1 | Mock engine + capability fixtures + browser-replay CUJ scripts (L1) | T0 | no | ?
| ? |

```

```

159 | T2 | Appliance shell: Caddyfile + systemd units; firewall; **edge auth**; rotate key
(P1) | ? | owner | ? | ? |
160 | T3 | L3 live posture test (raw `:8000`/`:900x` refused from outside) | T2 | no | ? | ? |
|
161 | T4 | S1 ingest (local upload wired) + segmenter picker | T1 | no | ? | ? |
162 | T5 | S2 progress (poll + one-job + restart-failure) | T4 | no | ? | ? |
163 | T6 | S3 embed **RallyPicker** + compile + download | T4 | no | ? | ? |
164 | T7 | *(engine)* `GET /api/capabilities` + `GET /api/health` + banner | T1 | coord | ?
| ? |
165 | T8 | *(engine)* `GET /api/videos` + `GET /api/reels/{video_id}` (S4 reload-survival) |
? | coord | ? | ? |
166 | T9 | S5 Settings (config+capabilities; disabled compute/storage target) | T7,T8 | no |
? | ? |
167 | T10 | Observability pass: correlation id end-to-end + structured logs + metrics (S5) |
T4 | no | ? | ? |
168 | T11 | Retention sweep for `output/+`uploads/` | ? | owner | ? | ? |
169 | T12 | Ingest from **bucket URI** (`s3://`/`gs://`) wiring ? **engine half DONE**
(`/api/process` accepts URIs, M2/[#280], `[verified]`); remaining = the SPA ingest call (=
**B-P2**) | T4 | coord | ? | engine done; SPA pending |
170 | T13 | **GPU host stand-up** (provider-agnostic) + wire `detector_impl=native`; owner
byte-parity (P4) | branch | ? owner | ? | ? |
171 | T14 | *(reserved)* Bucket-worker bridge ? remote-GPU "compute as a setting" (P5) |
T8,T12 | ? owner | ? | ? |
172
173 **Owner priority (2026-06-20): T13 (GPU host) is elevated to run in parallel from the
start** ? owner provisions the host + token; the engine session lands native WASB; the UI MVP
does **not** block on it (ships on Gemini).
174
175 ## 10. Open questions ? owner / external
176 **Blocking gates:** (1) **edge-auth** ? Caddy basicauth (self-contained) vs
cloudflared+Cloudflare Access (matches the two-host design, no inbound ports)? (2) **retention
defaults** ? auto-delete originals after compile? keep proxies+intervals only? window?
(privacy/consent). (3) **per-video Gemini/GPU cost ceiling** ? what cap, hard-fail vs warn? (4)
**GPU host choice** ? DO GPU droplet / GCE GPU VM / Cloud Run+GPU (and is a containerized worker
worth building for Cloud Run)? (5) **P5 bridge ownership** ? who builds the orchestration
endpoint + how it coordinates with the active worker sessions.
177 **Nice-to-knows:** console home (web/ vs engine static); a review-playback streaming
endpoint; per-user scoping timing; containerization vs runbook.
178
179 ## 11. Definition of done
180 - [ ] CUJ-1 works end to end on the droplet (Gemini **and** stub) behind the gate, via
the ingest menu.
181 - [ ] Raw `:8000`/`:900x` unreachable from outside (**tested**, not assumed).
182 - [ ] Local upload wired; bucket-URI ingest at least designed/tracked; Drive scoped
honestly (not faked).
183 - [ ] RallyPicker embedded with honesty rules; `min_shot_count` surfaced.
184 - [ ] `/api/capabilities` + `/api/health` + banner truthful; reload-survival via
`/api/videos` + `/api/reels`.
185 - [ ] **Observability:** correlation id threaded end-to-end; structured logs at every
new path; loud failures; logs inspected during CUJ replay.
186 - [ ] Retention sweep in place; `GEMINI_API_KEY` rotated; `allowed_video_root` set.
187 - [ ] CI green (L2 contract + L4 regression); coverage ? floor; S9 tracker all
?/?-resolved.
188
189 ## 12. References
190 **Engine code `[verified this session]`:**
`backend/main.py:12-17,42,52,307-328,331-340,342-396,638`;
`backend/api/uploads.py:64,89,123,147`; `backend/api/job_worker.py:37`;
`backend/api/static/app.js:88,344,373`;
`backend/storage/{base.py,local.py:33,ref.py,__init__.py:43}`;
`backend/config/models.py:129,183,298`;
`backend/pipeline/segmenters/{trajectory_hybrid.py:57-63,76-85, fusion_hybrid.py:76-86,
gemini.py:470-485, motion.py:191-198}`; `backend/pipeline/detectors/{native_wasb_runner.py,
stub_runner.py}`; `backend/worker/__main__.py:88,98,149`; `config.json` (`default_segmenter:
gemini`). **Engine docs:** `HOSTING_PLAN.md:42,102-108`; `VIDEO_LOCALITY_MODEL.md`;

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`DESKTOP_APP_PLAN.md`; `SETUP_NEW_MACHINE.md`; `PLATFORM_ARCHITECTURE.md`;
`DECOUPLED_COMPUTE/{README,02,04,07,HANDOFF}.md`; `UI_PROPOSAL.md:41-66`;
`PROJECT_DOC_TEMPLATE.md`. **This repo:** `PER_RALLY_SELECTION.md`, `REQUEST_ACCESS_FORM.md`,
`PORTABILITY.md`, `web/README.md`.
191
192 ---
193 ### Changelog
194 - 2026-06-20 ? Initial `DRAFT`. Consolidated three single-box-appliance designs
(FastAPI-native / unified-origin / decoupled-profile) into the reverse-proxied unified-origin
spine + compute-as-a-setting framing, via a multi-agent design pass; every engine-fact anchor
re-verified against source before writing (catching the native-WASB partial-wiring +
`ai_max_workers` default nuances). Folded in the owner's ingest-menu (Drive/bucket/local),
multi-provider GPU, and first-class Observability directives.
195

```

9. assistant (2026-06-21T13:56:32.635Z)

Now let me read the remaining docs: PER_RALLY_SELECTION.md, REQUEST_ACCESS_FORM.md, PORTABILITY.md, and NEXT_STEPS.md.

10. user (2026-06-21T13:56:33.822Z)

```

1      # Request-Access form ? segmented lead capture + setup discovery
2
3      > **Status:** `IN PROGRESS` . **Owner:** Avi . **Created:** 2026-06-20 . **Last
updated:** 2026-06-20
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived`
5      > **Relation:** first tracked work item in the `khelsutra` repo; mirrors the engine
repo's
6      > `PROJECT_DOC_TEMPLATE` discipline (status header . decisions-locked . $6 tracker .
DoD).
7      > Feeds the platform direction in the engine repo's `PLATFORM_ARCHITECTURE.md $3`
(StorageBackend /
8      > ComputeBackend) and `VIDEO_LOCALITY_MODEL.md` ? see $7.
9
10     ---
11
12     ## 0. TL;DR
13     Replace the landing page's `mailto:` "Request access" CTA with a real form that (a)
segments
14     **academies vs individual enthusiasts**, (b) asks the two questions that actually decide
how we'd
15     serve a user ? ***"do you have a GPU machine?"** (compute placement) and ***"where do your
videos
16     live?"** (storage backend) ? and (c) keeps a free-text message. Submissions are stored
in our own
17     **Cloudflare D1** via a small **Worker** (`POST /api/request-access`). The form doubles
as PMF +
18     architecture research.
19
20     ## 1. Problem & goal
21     Today the CTA is `mailto:hello@khelsutra.guru` ? zero structure, nothing collected, no
segmentation.
22     We want **structured, queryable** interest signals from two distinct audiences, captured
the moment
23     someone raises a hand, **without shipping lead data to a third party** (privacy-first
ethos).
24     **Goal:** a clean on-brand form whose responses land in our own Cloudflare account,
queryable later.
25
26     ## 2. Decisions locked
27     - **Backend = Cloudflare Worker + D1** (owner decision, 2026-06-20). Data stays in our
Cloudflare
28     account; queryable; no third party. (Alternatives considered: hosted form
(Tally/Formsfree ?

```

```

29     third-party data), Worker?email-only (no DB), structured `mailto` (no collection).
Rejected for
30     the reasons in parens.)
31     - **Two audience segments:** `academy` and `individual` ? the form lightly adapts
(academy also
32     asks club name + #courts).
33     - **The two discovery questions are first-class** (not optional polish): GPU/compute +
storage ?
34     they map 1:1 onto the BY0-compute / BY0-storage architecture forks (§7).
35     - **Free-text message retained** ("Anything else?").
36     - **Privacy:** store only what's submitted + a coarse country (from
`request.cf.country`); **no raw
37     IP**; consent checkbox for "email me about the alpha"; this data is
**operations/contact only**,
38     never a training source (same guardrail as the engine repo).
39     - **Deploy model changes:** the site moves from dashboard direct-upload ? **Wrangler**
(Worker with
40     static assets + D1 binding). Documented in §5; owner runs the `wrangler` deploy (our
account/auth).
41
42     ## 3. Form spec
43     | Field | Type | Notes |
44     |---|---|---|
45     | Segment | radio | `academy` \| `individual` (required) |
46     | Name | text | required |
47     | Email | email | required (so we can reply) |
48     | City | text | optional |
49     | Club / academy name | text | shown for `academy` |
50     | Courts recorded | select | `1` \| `2-4` \| `5+` (academy; optional) |
51     | **GPU machine for local processing?** | radio | `yes` \| `unsure` \| `no` \| `prefer-
cloud` |
52     | **Where do your recordings live / how would you share them?** | checkboxes (multi) |
`gdrive` \| `onedrive` \| `dropbox` \| `bucket` (S3/GCS) \| `local` (SD/drive) \| `unsure` |
53     | Message | textarea | free text ("Anything else?") |
54     | Consent | checkbox | "Okay to email me about the KhelSutra alpha" |
55     | `hp` | hidden honeypot | anti-spam; if filled ? silently drop |
56
57     ## 4. Design / architecture
58     - **Frontend** (`site/index.html`): the header + hero "Request access" buttons scroll to
the
59     `#join` form; `#join` becomes the form. On submit, JS `POST`s JSON to `/api/request-
access`.
60     - **Progressive enhancement:** if the endpoint isn't live yet (pre-Worker), it falls
back to a
61     prefilled `mailto:hello@` so the form is useful from the first re-upload.
62     - **Worker** (`/api/request-access`): validates (email present, honeypot empty), inserts
into D1,
63     returns `200 {ok:true}`; serves the static assets for all other routes.
64     - **D1 schema** (`access_requests`): `id, created_at, segment, name, email, city,
org_name, courts,
65     has_gpu, storage (csv), message, consent, country, user_agent`.
66     - **Anti-spam:** honeypot first; add **Cloudflare Turnstile** in P2 if spam appears.
67     - **Notify (P2):** optional email ping to `hello@` per submission (MailChannels /
Cloudflare Email).
68
69     ## 5. Deploy (owner-run; our Cloudflare account)
70     ```
71     wrangler d1 create khelsutra-access          # once; copy the database_id into
wrangler.toml
72     wrangler d1 execute khelsutra-access --file=site/schema.sql # create the table
73     wrangler deploy                               # ships the Worker + static assets + D1
binding
74     ```
75     > Until the Worker ships, the form works via the `mailto` fallback (re-upload `site/` as
today).

```

```

76      > Git?Cloudflare auto-deploy still intentionally not connected.
77
78      ## 6. Deliverables & progress tracker ? source of truth
79      Legend: ? Todo · ? In progress · ? Done. One PR per row off `origin/master`.
80      | ID | Deliverable | Gated? | Status | PR |
81      |----|-----|-----|-----|----|
82      | P0 | This design doc + `NEXT_STEPS.md` | No | ? | #2 |
83      | P1 | Form UI (segments + GPU + storage + free text + consent) w/ `mailto` fallback |
No | ? | #2 |
84      | P2 | Worker + D1 + `wrangler.toml` + `schema.sql` + **unit tests (10 green)** | owner
deploy | ? | this PR (built+tested; owner runs `wrangler d1 create` + `deploy`) |
85      | P3 | Anti-spam (Turnstile) + per-submission email notify | No | ? | ? |
86      | P4 | Tiny admin/export view (auth'd) to read responses | owner | ? | ? |
87
88      ## 7. Why this matters beyond lead capture
89      The two questions are **the architecture forks** from the engine repo's BYO analysis:
90      - **GPU?** ? compute placement: desktop/local (their GPU) vs hosted vs zero-infra-BYO
91      (`PLATFORM_ARCHITECTURE.md §3.2`, `DESKTOP_APP_PLAN.md` P0 spike).
92      - **Storage?** ? which `StorageBackend` + whether autosync is worth building
93      (`PLATFORM_ARCHITECTURE.md §3.1`, the cloud-storage/autosync thread).
94      So early responses directly inform what to build next ? not just who to email.
95
96      ## 8. Open questions
97      1. Turnstile now or wait for spam? (Default: honeypot now, Turnstile if needed.)
98      2. Email-notify per submission, or just query D1 weekly? (Default: notify in P2.)
99      3. Do we localize the form (Kannada/Hindi) for academies now or at beta? (Default:
English now.)
100
101      ## 9. Definition of done
102      - [ ] Form replaces the `mailto` CTA, segments academy/individual, asks GPU + storage +
free text.
103      - [ ] Submissions persist to Cloudflare D1 via the Worker; honeypot drops bots.
104      - [ ] Privacy posture honored (no raw IP; consent; contact-only).
105      - [ ] Deployed live on khelsutra.guru; a test submission appears in D1.
106      - [ ] Doc tracker + DoD updated; archived when P1?P3 land.
107

```

11. user (2026-06-21T13:56:34.122Z)

```

1      # Interactive per-rally selection ? reel builder (RallyPicker)
2
3      > **Status:** `IN PROGRESS` ? **MVP / M0 (P1?P5) COMPLETE 2026-06-20** (pick ? query ?
select ? compile ? download, on existing engine endpoints; verified e2e against a stub engine,
GCP-GPU validation next). Design **APPROVED** by owner 2026-06-20**. Downstream milestones M1?M5
(multi-video auth, per-rally preview endpoint, corpus loop, localized NL) remain owner-gated. ·
**Owner:** Avi · **Created:** 2026-06-20 · **Last updated:** 2026-06-20
4      > **Lifecycle:** `DRAFT ? IN PROGRESS ? DONE ? archived` (move to `docs/archives/` when
DONE)
5      > **Tracking anchors:** §7 progress tracker is the source of truth; indexed in
`docs/README.md`; pointer in `NEXT_STEPS.md` + memory `current-state` once work starts.
6      > **Relation:** second major tracked work item in the `khelsutra` repo (peer of
`REQUEST_ACCESS_FORM.md`); mirrors the engine repo's `PROJECT_DOC_TEMPLATE` discipline.
Primarily realizes alpha screen **A5 Highlights** (engine repo `UI_PROPOSAL.md:47`, exec-plan
**M4** `UI_ALPHA_EXECUTION_PLAN.md:39`); its correction-loop extension realizes **A4 Rally
Review** (`UI_PROPOSAL.md:46`, exec-plan **M3** `:38`). Consumes the engine's `/api/query` +
`/api/compile` contract.
7      > **Honesty note:** every engine-fact claim is tagged `[verified]` (read against engine
code this session), `[design]` (proposed here), or `[researched]` (needs sign-off). Not
legal/financial advice.
8
9      ---
10
11      ## 0. TL;DR
12      Build the **pick-rallies ? reel** surface as a **hybrid**: a selectable rally
**list/grid** is the workhorse; a **deterministic query bar** ("rallies over 10s, longest

```

first") pre-selects into the **same** selection set; a sticky footer compiles the selection into one downloadable MP4. The entire MVP runs on **existing engine endpoints** ? `POST /api/query` ? `POST /api/compile` ? `GET /api/highlights` `[verified backend/main.py:331-340, 342-396, 307-328]` ? with **zero new engine intelligence and zero new endpoints**. The single most important caveat: queries and chips are scoped to **length / count / order / shot-count only**; shot-type, player, and score are **roadmap-only** (no detector, no column) and must not be faked. The biggest UX-vs-effort tension: **no per-rally preview is possible today** ? only the **final compiled reel** can stream ? so MVP selects on text metadata, and visual preview/timeline is a deliberate later milestone.

13

14 **## 1. Problem & goal**

15 Today the alpha can detect rallies and stitch a reel, but reel-building is either a **silent `stitching.min\_shot\_count` threshold** `[verified backend/main.py:362-383]` or a static checkbox list whose data plumbing is even broken ? the bundled static UI reads `data.rallies` while the API returns `play\_segments`, so it renders nothing `[verified backend/api/static/app.js:344 vs backend/main.py:340]`. The owner's framing is **dynamic per-video selection, not a fixed threshold**: a coach should say "give me the long rallies from this session" and get exactly those, then fine-tune by hand.

16

17 **The concrete outcome ("good")**: A coach uploads a 40-minute session; the engine finds 60 rallies. They type **"top 15 longest"**, eyeball the auto-selected cards, untick two warm-up rallies, name it **"Tuesday ? long points"**, and download a ~6-minute reel ? without learning any threshold knob.

18

19 **Goal**: the smallest **honest** surface that turns engine-detected rallies into a user-curated, downloadable reel, with both **fast** (query) and **precise** (checkbox) selection sharing **one** state.

20

21 **Non-goals (alpha)**: no conversational LLM (`UI\_PROPOSAL.md:66` defers it); no shot-type/player/score filters (no data exists); no per-rally video preview while it requires unbuilt streaming; no multi-tenant auth (single-box alpha).

22

23 **## 2. Decisions locked**

24 | # | Decision | Source / date | Implication |

25 |---|-----|-----|-----|

26 | D1 | Dominant surface = selectable rally **list/grid** | Synthesis 2026-06-20 (Grid scored 15, fit-to-data 4) | Scales to long matches via bulk+filter; shows honest fields only |

27 | D2 | Fast entry = **deterministic** client-side `parseQuery()` (no LLM) | `[verified UI\_PROPOSAL.md:66]` conversational QA deferred | Predictable, unit-testable, offline, zero API cost |

28 | D3 | **One** shared `Set<segment\_id>`, mutated by query + checkbox + (future) timeline | Synthesis 2026-06-20 | No second compile path can diverge from what the user sees |

29 | D4 | Compile via existing `POST /api/compile {video\_id, segment\_ids, output\_filename}` | `[verified backend/main.py:342-396]` | Selection order is harmless ? stitcher `\_merge\_overlapping` de-dupes `[verified compiler.py:80-105, 311-317]` |

30 | D5 | Query grammar (alpha) = length, count, order, **shot_count only** | `[verified database.py:122-135, gemini.py:470-485]` | Only fields the payload actually returns; grey out the rest |

31 | D6 | MVP backend = **zero new endpoints** | `[verified]` reuse map (§4) | Ships the full loop on the already-shipped contract |

32 | D7 | Replace the silent `min\_shot\_count` cut with **explicit user selection** | Owner framing + `[verified main.py:362-383]` | UI **warns** about the floor; it never hides why a rally vanished |

33 | D8 | **Default selection = ON** (curate-by-removal) | Owner 2026-06-20 | Rallies load all-selected (matches today's "all rallies" reel); user removes warm-ups. Footer/empty UX assumes a non-empty start |

34 | D9 | **Preview-before-select** is NOT a ship-blocker for the MVP | Owner 2026-06-20 | MVP selects on text metadata + previews the **final compiled reel**; the source/clip-stream endpoint (per-rally preview) is the next milestone (P9), not MVP |

35 | D10 | **MVP is job-tethered** (carry `result.video\_id`); no `GET /api/videos` | Owner 2026-06-20 | Smallest honest slice; multi-video home + identity scoping is P8 (gated) |

36 | D11 | `min\_shot\_count`: **warn, do NOT override** in the MVP | Owner 2026-06-20 | The UI surfaces the floor; an explicit-selection override of the compile default is deferred (would trigger the Launch Report convention) |

37

```

38     ## 3. Foundation ? what the engine gives us today `[verified this session]`
39     The whole design rests on the engine's real, already-shipped contract. Read against the
engine repo (`badminton-highlight-indexer`):
40
41     - The rally store is `play_segments` ? columns `id, video_id, start_time, end_time,
duration, server, receiver, metadata` `[verified database.py:124-135]`. `id` is the integer the
UI selects and passes to compile; `start_time/end_time/duration` are REAL seconds (duration
computed at insert).
42     - Listing rallies = `POST /api/query {video_id}` ? `{play_segments:[...]}` ? the
*only* rally-list path today; it JSON-parses `metadata` and joins `events` `[verified
main.py:331-340, database.py:427-475]`.
43     - Building a reel = `POST /api/compile {video_id, segment_ids:[int],
output_filename}` ? loads `videos.filepath`, filters all segments to the requested
`segment_ids`, applies the `min_shot_count` floor, extracts `time_pairs=[(start,end)]`, and
calls `VideoCompiler.compile_highlights(raw_path, segments, name)` `[verified main.py:342-396,
compiler.py:303]`, which merges overlaps then per-clip re-encodes ? concat ? one MP4
[verified compiler.py:311-317]. Selection order/duplicates are therefore safe.
44     - Downloading = `GET /api/highlights/{video_id}/{filename}` ? streams the compiled
MP4. Compile returns a server-local filepath, NOT a URL [verified main.py:394], so the SPA
constructs this download URL itself. The compiled reel is the only streamable video that
exists today [verified main.py:307-328].
45     - The floor knob is readable ? `GET /api/config` exposes `stitching.min_shot_count`
[verified main.py:121-123], so the UI can warn before a selected rally is silently dropped.
46     - Job entry ? `GET /api/jobs/{job_id}` returns `result.video_id` [verified
main.py:274-293], which is how the MVP reaches a video without a (nonexistent) list-videos
endpoint.
47
48     Honest fields vs fabricated/roadmap (critical for the UI):
49     - Honest today (default `default_segmenter: "gemini"`, per `config.json`
[verified]): `start_time`, `end_time`, `duration`, and `metadata.shot_count` (Gemini's
`shuttle_exchange_count`, fallback `max(3, int(dur/0.8))`) + `metadata.shot_count_confidence` (a
string, often "Unknown") [verified gemini.py:470-485]. On the gemini path
`server`/`receiver` are written `None` ("no fabricated data") [verified gemini.py:480-487].
50     - Fabricated ? do NOT surface: the `motion` demo segmenter fills `server`/`receiver`
via `random.sample(...)` , a random `shot_count`, and random `events` (serve/smash/foul/winner)
[verified motion.py:191-198]. `query` returns these columns + `events`, but they are not
ground truth and must stay hidden.
51     - Roadmap ? no column, no detector: shot-type per shot, player identity, match/point
score, game/set grouping. None exist anywhere in the schema; do not consume them [verified
database.py whole-schema 122-148].
52     - Correction-loop columns exist but are dead:
`candidate_segments.human_label/notes/confirmed_segment_id` are defined but no endpoint writes
them [verified database.py:166-168] ? that's the M4 corpus extension (engine PR-4/B6,
`UI_ALPHA_EXECUTION_PLAN.md`).
53
54     ## 4. Design / architecture
55
56     Reuse map (existing ? [verified], no change for MVP): POST /api/query` (list) ?
POST /api/compile` (stitch) ? GET /api/highlights/{video_id}/{filename}` (download); GET
/api/config` (floor) ; GET /api/jobs/{id}` (carry `result.video_id`). All net-new code is the
SPA in `khelsutra` `web/` (Vite + React + TS + Tailwind, per `web/README.md`), plus a small
set of optional real-gap engine enablers (§4.3) ? none required for MVP.
57
58     4.1 Components (net-new SPA)
59     - QueryBar + `parseQuery()` ? a *pure, deterministic, unit-tested* string ?
`{filter, sort}` compiler over `duration` / `ordinal` / `shot_count`. Renders a *parsed-pill*
("filter: duration > 10s · sort: duration desc") so the interpretation is transparent and hand-
correctable; greys out shot-type/player/score tokens with an inline "not available yet". No LLM,
no network call.
60     - RallyList/Grid ? one card per `play_segment`: ordinal (by `start_time`), `mm:ss`
start, duration (`end?start`), and a shot-count + confidence chip only when
`metadata.shot_count` is present (absent on detector paths that don't write it). Checkbox per
card; the thumbnail area is a placeholder (no endpoint yet).
61     - useRallySelection() hook ? owns `selectedIds:Set<number>`, `displayOrder`,
`apply(queryResult)`, `toggle(id)`, and bulk Select-all / Clear / Invert. Unit-tested (matches

```



```

the exec-plan's "API client + review-state logic" bar).
62     - **SelectionFooter (sticky)** ? live "N clips / total M:SS"; a **`min_shot_count`
warning** when a selected rally would be dropped at compile (read from `GET /api/config`); reel-
name input; one **Build** button.
63     - **CompileResult** ? `<video src=GET /api/highlights/...>` preview of the *compiled
reel* + a download link.
64     - **TypedApiClient** ? `query`, `compile`, `buildDownloadUrl`; the typed seam to the
engine.
65
66     **4.2 Data flow** `[verified anchors inline]`
67     Gemini segmenter (process time) writes `play_segments {id, start/end_time, duration,
metadata.shot_count/confidence/detector}` `[geminipy:461-498]`
68     ? SPA `POST /api/query {video_id}` ? `{play_segments:[...]}` `[main.py:331-340]`
(**fix** the static UI's `data.rallies` read ? `play_segments` `[app.js:344]`)
69     ? normalize to `RallyCard[]` + one shared `selectedIds:Set`;
`server`/`receiver`/`events` are present in the payload but motion-only fabrications ? **not
surfaced** `[motion.py:191-198]`
70     ? all input modes mutate the **same** `selectedIds`: (a) `parseQuery()` predicate over
duration/ordinal/shot_count; (b) checkbox toggle + bulk ops; (c) *(post-MVP)* timeline band
71     ? footer recomputes count + ?duration **client-side**, and warns for any selected rally
with `shot_count < stitching.min_shot_count` `[main.py:362-383]`
72     ? **Build** ? `POST /api/compile {video_id, segment_ids: [...selectedIds],
output_filename}` ? stitcher de-dupes via `_merge_overlapping` `[compiler.py:311-317]`, cuts
`time_pairs` from `videos.filepath`, re-encodes per clip ? concat ? one MP4 `[main.py:386-393,
compiler.py:303]`
73     ? compile returns a **server-local filepath, not a URL** `[main.py:394]` ? SPA builds
`GET /api/highlights/{video_id}/{output_filename}` to preview in `<video>` + download
`[main.py:307-328]`.
74
75     **4.3 Engine API additions (none for MVP; tagged against real gaps)**
76     | Endpoint | Purpose | Status |
77     |---|---|---|
78     | `POST /api/query` | List rallies (only path) ? fix SPA to read `play_segments` |
**exists** |
79     | `POST /api/compile` | Stitch selected `segment_ids` ? MP4 | **exists** |
80     | `GET /api/highlights/{video_id}/{filename}` | Stream/download compiled reel |
**exists** |
81     | `GET /api/config` | Read `stitching.min_shot_count` for the warning | **exists** |
82     | `GET /api/jobs/{job_id}` | Carry `result.video_id` into the picker | **exists** |
83     | `POST /api/compile` ? add `download_url` | Return `/api/highlights/...` so SPA stops
hand-building it (additive, low-risk; today returns only a filepath `main.py:394`) | **modify**
|
84     | `GET /api/health` | Engine-offline banner ping the SPA is specified to use
(`HOSTING_PLAN.md:42`) | **new** |
85     | `GET /api/videos/{video_id}/rallies` | RESTful/cacheable alias returning the same
`{play_segments}` shape | **new** |
86     | `GET /api/videos` | List a user's videos for a Match picker (only `get_video(id)`
exists, `database.py:418`) ? needs DB enumerate + identity scoping | **new** |
87     | `GET /api/videos/{video_id}/source` (Range) *or* `/clip?start=&end=` | Stream
source/per-span video so a rally previews before selection ? neither exists | **new** |
88     | `GET /api/videos/{video_id}/rallies/{segment_id}/thumbnail` | Per-rally poster still
(no thumbnail column; WASB frames are transient) | **new** |
89     | `PATCH /api/segments/{id}` | Persist accept/reject/nudge ?
`candidate_segments.human_label/notes/...` (columns exist, unwritten ? corpus loop) | **new** |
90
91     **? Dead-ends / do-NOT:** never surface `server`/`receiver`/`events` (motion-only
randoms `[motion.py:191-198]`); never consume shot-type/player/score (no data); never render a
confidence **meter** ? `shot_count_confidence` is a Gemini *string* (often `Unknown`) and
`play_segments` has **no** numeric confidence column.
92
93     **4.4 Observability (first-class, per `[memory: feedback-launch-quality-bar]`)** Every
P3 user action emits a **structured JSON log line** (`web/src/lib/log.ts`) so a journey is
traceable from text to selection ? the SPA-side "runlog". Each query application generates **one
`cid` shared by every line of that journey; the future `/api/compile` call (P5) reuses the
same `cid` to thread the trace into the engine. Events (all on the P3 path):

```

```

94
95     | event | level | when | key fields |
96     |---|---|---|---|
97     | `query.parse` | info | every apply attempt | `cid`, `source` (typed/example),
`recognized`, `filters`, `sort`, `limit`, `unavailable` ? the exact interpretation |
98     | `query.unavailable_request` | info | the query names a shot-type/player/score | `cid`,
`raw`, `unavailable[]` ? **demand signal** for roadmap features (what users ask for that we
can't yet do) |
99     | `query.skip` | info | nothing actionable parsed | `cid`, `raw`, `reason` |
100    | `query.apply` | info | a recognized query is applied | `cid`, counts
(`total`/`selected`), `selected_ids`, `dropped_unavailable` |
101    | `query.empty_result` | **warn** | recognized query matches **zero** rallies | `cid` +
full interpretation ? a looks-empty grid is **never silent** |
102    | `query.apply_failed` | **error** | the (pure) parser/selector unexpectedly throws |
`cid`, `raw`, `error` ? loud, never swallowed |
103    | `selection.bulk` | info | Select-all / Clear / Invert | `cid`, `op`, `result_count` |
104    | `rally.toggle` | debug | a card is added/removed | `id`, `action` |
105
106    No silent drops: unsupported concepts (`query.unavailable_request` +
`dropped_unavailable`) and empty results (`query.empty_result` warn) are always logged. Verified
live in-browser (events fire with correct `cid`/fields, zero console errors).
107
108    ## 5. Risk table
109    | Risk | Severity | Mitigation |
110    |---|---|---|
111    | **No source/clip/frame/thumbnail endpoint** ? only compiled reels stream
`[main.py:307-328]` | High (UX) | MVP selects on text metadata + previews the final reel;
preview/timeline deferred to M3 behind a new endpoint |
112    | `min_shot_count` silently drops selected rallies `[main.py:362-383]` | High (looks
broken) | Footer warns *before* Build; D7 makes selection explicit; an override is an owner gate
(Launch-Report-worthy `[memory: launch-report-convention]` ) |
113    | No match-duration field ? a timeline must derive `total = max(end_time)`, mis-
rendering pre/post dead air | Med | Timeline deferred to M3; never over-promise it as a true
scrubber until source streams |
114    | `shot_count` is an *estimate* (`shuttle_exchange_count` w/ `max(3,int(dur/0.8))`
fallback) `[geminipy:470-485]` | Med | Label "approx."; confidence shown as the raw string, not
a meter |
115    | No auth / user scoping anywhere (CORS `*`, `videos.user_id` never set/filtered) | High
for multi-tester | Fine for single-box local alpha; a ship-blocker for any shared host ? gate
`GET /api/videos` behind PR-3 scoping |
116    | compile returns a filepath not a URL `[main.py:394]` | Low | Client builds the URL for
MVP; `download_url` field (M1) removes the brittleness |
117    | Surface sprawl (query + grid + future timeline + future correction) | Med | Deliberate
IA: query+grid = one screen; correction = separate ? corpus screen (owner call, S8) |
118
119    ## 6. Honest limits ? what this does NOT do
120    A green SPA build + passing `parseQuery`/selection unit tests prove the **picker logic
only** ? **not** that the engine's rally boundaries are correct (the measured **rally-END**
weakness is a separate quality track, unaffected by this UI). "Conversational" here means a
**deterministic four-dimension parser**, not natural-language understanding. Nothing is
implemented in `web/` yet ? this is `[design]`. The feature curates *what the engine already
found*; it cannot surface a rally the detector missed, nor split/merge a mis-bounded one (that's
the M4 correction loop).
121
122    ## 7. Deliverables & progress tracker ? **source of truth**
123    Legend: ? Todo · ? In progress · ? Done · ? Blocked/gated. **One small PR per row**,
branched from `origin/master`, no stacking. CI (`npm test`) must stay green; merge to `master`
auto-deploys the site (docs/`web/` changes don't touch the live `site/` Worker).
124
125    | ID | Deliverable | Depends on | Gated? | Status | PR |
126    |----|-----|-----|-----|-----|----|
127    | P0 | This design doc + `docs/README.md` + `NEXT_STEPS.md` pointers | ? | No | ? |
[#11](https://github.com/avidullu/khelsutra/pull/11) |
128    | P1 | SPA scaffold (Vite+React+TS+Tailwind) in `web/` + typed API client; dev proxy to
engine `:8000` | P0 | No | ? | [#14](https://github.com/avidullu/khelsutra/pull/14) |

```

```

129 | P2 | RallyList/Grid (honest fields only) + `useRallySelection()` hook (unit-tested) |
P1 | No | ? | [#21](https://github.com/avidullu/khelsutra/pull/21) |
130 | P3 | `parseQuery()` + parsed-pill + example chips (unit-tested; greys out unavailable
tokens) | P2 | No | ? | (this PR) |
131 | P4 | SelectionFooter + `min_shot_count` warning (from `GET /api/config`) | P2 | No | ?
| (this PR) |
132 | P5 | Build ? `POST /api/compile` ? client-built download URL + `` preview of
the reel | P2 | No | ? | (this PR) |
133 | P6 | *(engine repo)* `GET /api/health` + offline banner; `/api/compile` returns
`download_url` | ? | No | ? | ? |
134 | P7 | *(engine repo)* `GET /api/videos/{id}/rallies` RESTful alias | ? | No | ? | ? |
135 | P8 | *(engine repo)* `GET /api/videos` + PR-3 identity scoping ? A1 Matches screen |
P7 | ? owner | ? | ? |
136 | P9 | *(engine repo)* source/clip stream + timeline + per-rally preview/thumbnail | P8
| ? owner | ? | ? |
137 | P10 | *(engine repo)* `PATCH /api/segments/{id}` corpus loop (PR-4/B6) ? A4 Rally
Review | P9 | ? owner | ? | ? |
138
139 **Milestone rollout:** *M0/MVP = P1?P5 ? COMPLETE (2026-06-20)** (pick?reel on existing
API, zero new engine endpoints) . **M1 = P6?P7** (RESTful + reliability polish) . **M2 = P8**
(multi-video home) . **M3 = P9** (per-rally preview/timeline) . **M4 = P10** (corpus correction
loop) . **M5** = swap `parseQuery()` for localized NL behind the same seam (gated on shot/player
data landing).
140
141 ## 8. Open questions ? owner / external
142 **Blocking gates ? ? RESOLVED 2026-06-20 (owner) ? see D8?D11 in §2:**
143 1. ~~Default selection ON vs OFF?~~ ? **ON / curate-by-removal** (D8).
144 2. ~~Is preview-before-select a ship-blocker?~~ ? **No** ; text-metadata MVP, preview is
P9 (D9).
145 3. ~~Job-tethered vs multi-video MVP?~~ ? **Job-tethered** (D10).
146 4. ~~`min_shot_count` override?~~ ? **Warn, don't override** in MVP (D11).
147
148 **Still open (nice-to-knows, non-blocking):**
149 5. May user-facing copy call `parseQuery()` "conversational", given conversational QA is
explicitly deferred (`UI_PROPOSAL.md:66`)? Risk of over-promising NL. *Owner positioning/honesty
call.* *(Lean: avoid the word; show the parsed-pill instead.)*
150 6. One screen for query+grid (recommended) vs a separate ? corpus/correction screen for
A4? *Owner positioning call.*
151
152 ## 9. Definition of done
153 - [ ] From a finished job, a user can list rallies, query/select, and download a reel ?
on **existing** endpoints.
154 - [ ] Only honest fields (ordinal / duration / shot_count+confidence-when-present) are
shown; fabricated `server`/`receiver`/`events` are hidden.
155 - [x] `parseQuery()` covers length/count/order/shot_count, greys out unavailable tokens,
shows a visible parsed-pill, and is unit-tested. *(P3)*
156 - [x] The P3 path is **observable**: structured, correlation-id'd logs for parse / apply
/ skip / empty / unavailable-request / manual-edit; unsupported concepts and zero-result queries
are logged loudly, never dropped silently (§4.4). *(P3)*
157 - [x] Selection hook is unit-tested; the `min_shot_count` floor is surfaced, not hidden.
*(P2 hook . P4 floor warning ? `lib/floor.ts` mirrors `main.py:362-383`, exempts no-shot_count
rallies, blocks Build when all-dropped)*
158 - [x] From a finished job, a user can list rallies, query/select, and download a reel ?
on **existing** endpoints. *(P5 ? `?video=<id>` loads via `POST /api/query`; Build ? `POST
/api/compile` ? `` preview + download over `GET /api/highlights/...`; verified e2e
against a stub engine)*
159 - [x] `npm run build` (and `npm test`) clean; the static UI's `data.rallies` bug is gone
? the SPA **supersedes** the static `app.js` for this flow (reads `play_segments`).
160 - [x] §7 tracker + this DoD updated. *(Doc stays live for the owner-gated M1?M5 engine
milestones; not archived ? only the MVP/M0 slice is complete.)*
161
162 ## 10. References
163 **Engine code anchors [verified this session]:** `backend/main.py:121-123` (config),
`:274-293` (jobs), `:307-328` (highlights download), `:331-340` (query), `:342-396` (compile +
`min_shot_count` floor + filepath-not-URL); `backend/infrastructure/database.py:124-135`

```

```
(play_segments), `:166-168` (dead correction columns), `:418` (no list-videos), `:427-475`
(query read path + events join); `backend/pipeline/segmenters/gemini.py:461-498, 470-485`
(honest shot_count, server/receiver=None); `backend/pipeline/segmenters/motion.py:191-198`
(fabricated fields); `backend/pipeline/stitcher/compiler.py:80-105, 303, 311-317` (merge/de-dupe
+ entryptoint); `backend/api/static/app.js:344` (`data.rallies` bug); `config.json`
(`default_segmenter: "gemini"`).
164    **Engine docs:** `UI_PROPOSAL.md:46-47, 66`; `UI_ALPHA_EXECUTION_PLAN.md:38-39`;
`HOSTING_PLAN.md:42`; `OWNED_MODEL_IMPLEMENTATION_PLAN.md:54-55` (conversational seam built now,
model punted). **This repo:** `web/README.md`, `REQUEST_ACCESS_FORM.md`.
165
166    ---
167    ### Changelog
168    - 2026-06-20 ? Initial `DRAFT`. Consolidated three candidate designs (Timeline / Grid /
RallyPicker) into the hybrid recommendation via a multi-agent design pass; every engine-fact
anchor re-verified against source before writing.
169    - 2026-06-20 ? Owner **approved** the recommendations; locked D8?D11 (default ON .
preview not a ship-blocker . job-tethered MVP . warn-don't-override the floor); status `DRAFT ?
IN PROGRESS`; P0 done.
170    - 2026-06-20 ? **UI: lead with rally LENGTH, park shot-count as experimental** (owner).
Rally cards emphasize duration (`m:ss long`); the shot-count chip is flagged amber
"experimental" with a tooltip ("under development, may be inaccurate ? filter by length
instead"); example chips are duration-led (dropped "at least 8 shots"); a note states "rally
length is fully supported, shot-count is experimental". `parseQuery` still accepts shot-count
queries (not broken) but they're de-featured. Public badge `P5`?`alpha`. A test pins that every
example chip parses to a recognized, fully-available (non-shot-count) query. Suite 75 green.
171    - 2026-06-20 ? **P5 shipped ? MVP / M0 COMPLETE.** `App` now loads a real session from
`?video=<id>` (`POST /api/query`, default ALL-selected D8) with a mode banner (demo / loading /
loaded N / error); **Build** ? `compileReel` (`POST /api/compile`) ? `CompileResult` previews
the stitched reel in `<video>` over `GET /api/highlights/{video_id}/{filename}` (client-built
URL ? engine returns a filepath, not a URL) + a download link; `reelFilename()` slugifies to an
engine-safe bare `.mp4`. Loading/building/done/error states, all traced (`rallies.load_`,
`build.start/done/failed`). Verified e2e against a stub engine: load 5 rallies ? real
`/api/config` floor warning ? Build ? `<video>` reached `readyState=4` (decoded a real mp4
through the proxy) + download link correct; all-below-floor selection keeps Build disabled (P4
guard prevents the 400). `reelFilename` unit-tested; suite **73 green**, typecheck + build
clean. **Next: GCP-GPU end-to-end validation (real video ? rallies ? reels).**
172    - 2026-06-20 ? **P4 shipped**: `SelectionFooter` (live summary + reel-name input +
Build) with an **honest `min_shot_count` warning** ? `lib/floor.ts` mirrors the engine floor
(`main.py:362-383`) EXACTLY: a rally is dropped only if it has a numeric `shot_count` below the
floor; rallies *without* `shot_count` are exempt; **all-below ? Build disabled** (the engine
would 400). Config read from `GET /api/config` on mount (graceful offline). 8 floor unit tests;
amber/red warning states verified live in-browser; suite **70 green**, typecheck + build clean.
173    - 2026-06-20 ? **P3 shipped**: deterministic `parseQuery()` (`web/src/lib/query.ts`)
over length/count/order/shot_count + `runQuery`/`sortRallies`/label helpers and the `QueryBar`
(live parsed-pill, example chips, greyed "not available yet" for shot-type/player/score ? never
faked). Writes the shared selection set via a new `useRallySelection().apply`; `RallyGrid`
ordinals made stable so a re-sort doesn't renumber cards. **First-class observability ($4.4):**
structured, correlation-id'd `query.*` / `selection.*` / `rally.toggle` runlog (loud
`empty_result`/`apply_failed`; `unavailable_request` demand signal). **Adversarial multi-agent
review** (grammar/honesty/robustness/reuse) ran pre-merge ? 13 confirmed findings folded in,
incl. 4 majors: an honesty leak (`at least 8 smashes` ? fake `length` ? 8s), a selection-wiping
`top 0`, a unit-blind `first 30 seconds`, and self-contradictory pill labels (`first 10` shown
as "top 10"). Live browser CUJ replayed (filter / top-N / honest-greying / untick / no-op /
wipe-closed). Suite **62 green**, typecheck + build clean.
174
```

12. assistant (2026-06-21T13:56:36.938Z)

Now let me read the remaining two docs: PORTABILITY.md and NEXT_STEPS.md.

13. user (2026-06-21T13:56:37.886Z)

```
1    # Portability ? Cloudflare now, always-on VM if needed
2
3    > **Status:** design + proven (2026-06-20). **Decision:** keep the site/form host-
```

```

portable so
4      > moving off Cloudflare to an owner-run always-on VM ("if Cloudflare acts out") is an
**adapter
5      > swap, not a rewrite.**
6
7      ## The shape that makes it portable
8      - **Logic is runtime-agnostic.** `src/index.js` uses Web-standard `Request`/`Response`
and pure
9      helpers (`validate`, `toRow`, `isBot`, `handleRequest`) ? no host-specific calls.
10     - **All DB coupling is in adapters** (`src/store.js`): `d1Store` (Cloudflare D1) and
`sqliteStore`
11     (node:sqlite / better-sqlite3). `handleRequest(request, store)` takes the store by
injection.
12     - **One schema, one SQL.** `schema.sql` and the `INSERT` are plain SQLite, so D1 and a
VM's SQLite
13     file are byte-for-byte compatible ? zero dialect change. (Postgres later = minor
placeholder tweaks.)
14
15     ## Two hosts, same code
16     | | Cloudflare (now) | Always-on VM (fallback) |
17     |---|---|---|
18     | Entry | `src/index.js` (Worker `fetch`) | `server.js` (node:http) |
19     | Static | `env.ASSETS` | `server.js` serves `public/` |
20     | Storage | `d1Store(env.DB)` ? D1 | `sqliteStore(db)` ? node:sqlite file |
21     | Country | `cf-ipcountry` header | optional (GeoIP / null) |
22     | Deploy | git ? Workers Builds | `node server.js` under systemd |
23     | TLS/DNS | Cloudflare | keep Cloudflare in front, **or** Caddy/Let's Encrypt |
24
25     **Enforced, not asserted.** `node --test src/` runs BOTH paths and CI
(`.github/workflows/test.yml`,
26     Node 24) runs them on every PR, so the repo stays compatible with both hosts:
27     - `src/index.test.js` ? the Cloudflare Worker + D1 logic.
28     - `src/server.test.js` ? **boots the real `server.js`** (node:http + node:sqlite) and
round-trips it
29     over HTTP into SQLite (the same proof we run on the droplet, automated).
30
31     For a live check against a real VM: `SSH_TARGET=root@host scripts/droplet-smoke.sh` (scp
+ boot +
32     round-trip + cleanup). Verified live on the owner's droplet (Ubuntu 24.04, isolated
`/srv/khelsutra-site`).
33
34     ## Provider-portable provisioning ? destroy ? resume anywhere (scripted + TESTED)
35     Spinning up a fresh VM on **any** provider (DigitalOcean / GCE / any Ubuntu box) and
tearing it down
36     without losing data is a **first-class, scripted** flow ? so destroying a box and
resuming elsewhere is
37     routine, not risky.
38
39     - **`scripts/provision-vm.sh`** ? run from your LOCAL machine against a FRESH Ubuntu VM.
Pushes `site/`
40     via `git archive` (**private-repo-safe** ? no clone, token, or secret on the VM),
installs Node (?24,
41     so `node:sqlite` is unflagged), installs + starts a **systemd** service, opens the
firewall, and
42     **verifies** it serves (form + a real POST round-trip). Idempotent.
43     ...
44     SSH_TARGET=root@<ip> [SSH_KEY=~/.ssh/khelsutra_droplet] [PORT=8787]
[RESTORE_DB=./backup.db] \
45     site/scripts/provision-vm.sh
46     ...
47     - **`scripts/teardown-vm.sh`** ? **exports the SQLite state DB to a local timestamped
backup BEFORE**
48     anything is destroyed (registrations survive the move), stops the service, and
optionally destroys the
49     DO droplet (`DESTROY=1 DROPLET_ID=<id>`, via `doctl`).

```

```

50     ``
51     SSH_TARGET=root@<ip> [BACKUP_DIR=./vm-state-backups] [DESTROY=1 DROPLET_ID=<id>] \
52     site/scripts/teardown-vm.sh
53     ``
54     - **Clean resume on a different provider:** `teardown-vm.sh` (export) ? destroy ?
`RESTORE_DB=<backup>
55     SSH_TARGET=root@<newip> site/scripts/provision-vm.sh`. The restored DB is `scp`'d in
before the service starts.
56
57     **Verified live (2026-06-20):** a fresh D0 Ubuntu-24.04 droplet went bare ? operational
+ publicly
58     serving (HTTP 200) via `provision-vm.sh` (Node 24 auto-installed, systemd unit,
firewall); `teardown-vm.sh`
59     exported the state DB (12 KB); a re-provision with `RESTORE_DB` preserved the
registration (`rows:2` =
60     restored + the verify POST) ? then the throwaway droplet was destroyed. The
destroy?resume cycle is
61     reproducible across providers.
62
63     ## VM deploy runbook ("if Cloudflare acts out")
64     Node 22+ only ? **no `npm install`** (node:sqlite is built in).
65     ``
66     # on the VM:
67     git clone ? && cd khelsutra/site          # or scp the site/ folder
68     PORT=8080 DB_PATH=/var/lib/khelsutra/access.db node server.js
69     # put it under systemd (auto-restart) + a reverse proxy (Caddy/nginx) for TLS,
70     # then point DNS at the VM. ~1 hour; logic + schema unchanged.
71     ``
72
73     ## Honest notes
74     - "Cloudflare acts out" usually means **Workers/D1 pricing or limits, not DNS** ? so the
cheapest
75     move is to keep Cloudflare DNS/CDN in front and shift only the **compute** to the VM
(point a
76     route/origin at the box). Full exit (own DNS + Caddy TLS) is also supported.
77     - You can run **both** (Cloudflare primary + VM warm standby) since the schema is
identical; a
78     periodic `access_requests` export keeps them in sync.
79     - `server.js` is intentionally dependency-free; if you later want Postgres or a
framework (Hono runs
80     on Workers *and* Node), it slots in behind the same `store` + handler.
81

```

14. user (2026-06-21T13:56:38.055Z)

```

1     # Next steps (live) ? KhelSutra brand / site
2
3     Prioritized "what to do next" for the **`khelsutra` product repo**. Refresh at each
checkpoint.
4
5     **Where we are (2026-06-20):** `khelsutra.guru` is **LIVE and fully shipped** ? landing
page + `/capture`
6     checklist + OG image + a **segmented Request-Access form** (academy/individual + GPU +
storage + free
7     text). On **Cloudflare** (Worker `hidden-king-khelsutra` + D1 `khelsutra-access`): the
form persists to
8     D1 **and emails a summary to the owner on every registration** (Email Routing send
binding ? verified
9     live). **Git auto-deploy is connected** (Workers Builds, root `site/`): merge to
`master` ? deploys;
10    **CI runs `npm test` (20 green) on every PR**. The same form also runs on a dependency-
free **Node +
11    SQLite** host (`site/server.js`), **live + browsable on the D0 droplet ?
http://168.144.126.176:8787**
12    (systemd `khelsutra-site`). PRs #1?#8 merged. Docs: `docs/REQUEST_ACCESS_FORM.md`,

```

```

`docs/PORTABILITY.md`.
13
14     ## Next product threads (owner ideas, 2026-06-20)
15     - **? ACTIVE FOCUS (owner directive 2026-06-21): Friendly & Multilingual** ? make the
product
16     **super user-friendly for non-tech-savvy** coaches/parents **and localize it in Hindi
+ Kannada**.
17     **? Tracked design + P-tracker (source of truth): `docs/FRIENDLY_AND_MULTILINGUAL.md`
(DRAFT,
18     owner-gated).** Both surfaces have zero i18n today; the foundation (shared
`locales/{en,kn,hi}`
19     catalogs + negotiation contract + CI parity) is the deliverable. **Recommended first
slice =
20     SPA i18n scaffold + language switcher (en-only), unblocked.** 5 open owner decisions
in §8
21     (font OFL carve-out · en+kn-then-hi · backend depth · surface-first · catalog home).
Adapts the
22     engine's `I18N_PLAN.md` (P1?`site/` shim, P3?`web/` react-i18next) + the UX half of
`UI_PROPOSAL.md`.
23     - **Marketing video on the site** ? embed a short walkthrough of the record ? index ?
reel process,
24     themed for **serious enthusiasts**, on the landing page. ? **A real-UI CUJ screencast
now exists**
25     (Playwright recording of the per-rally SPA driven against real GPU-run data ?
`promo_cuj.mp4/`.gif`
26     in `gs://kheIsutra-rally-corpus/promo/`). Decide: use it as the marketing clip (?
shows real court
27     footage + a dev "P5" badge ? clean up before public embed), and/or produce a polished
Grok/Gemini cut.
28     - **Interactive per-rally selection** (big product feature) ? ? **MVP / M0 COMPLETE**
(the full
29     pick ? query ? select ? compile ? download loop on existing engine endpoints). Lives
in the alpha SPA
30     (`web/`). **? Tracked design + P-tracker (source of truth):
`docs/PER_RALLY_SELECTION.md` (IN PROGRESS;
31     owner-approved, D8?D11 locked).** **Done: P0 design (#11), P1 scaffold (#14), P2
grid+hook (#21), P3
32     `parseQuery()` query bar (#28), P4 SelectionFooter + `min_shot_count` warning (#30),
P5 real load +
33     Build?compile?preview?download (#31).** Verified e2e vs a stub engine + a CUJ replay
against the real
34     L4-run rallies + reel. **NEXT (owner-gated, engine repo): M1?M5** ? `GET /api/health`
+ `download_url`
35     (P6), RESTful `~/rallies` alias (P7), multi-video home + auth (P8), per-rally
preview/timeline (P9),
36     corpus correction loop (P10), localized NL (M5). **Optional:** a fresh live
`/api/process` GPU
37     inference once the engine session's M2 training frees the single GPU.
38
39     ## Queued (site polish)
40     - **Always-Use-HTTPS + HSTS** (Cloudflare SSL/TLS) ? closes the plain-HTTP flag.
41     - **Cloudflare Web Analytics** ? paste the token into `site/public/index.html` (or
enable auto-setup).
42     - **`/capture` footer link** ? currently `capture.html` (307?`capture`); change to
`capture`.
43     - **Droplet ? production** ? TLS + a subdomain (`vm.kheIsutra.guru` via Caddy) if it
should be a real
44     standby; add **VM-path SMTP** so the Node host also emails (currently `noopNotify`).
45     - **DMARC** TXT for deliverability; delete any orphan `kheIsutra-site` Worker; clean CF
D1 test rows
46     (`deploy-test@`, `email-test@`). Revoke the droplet SSH key
(`~/ssh/kheIsutra_droplet`) when done.
47
48     ## Bigger arcs (design-first)
49     - **Single-box appliance (KheIsutra Studio)** ? **? Tracked design:

```

```

`docs/LOCAL_APPLIANCE_ARCHITECTURE.md` (DRAFT, owner-gated).**
50     The `web/` SPA drives the full **ingest (Drive / bucket URI / local upload) ? index ?
pick rallies ? reel ? download**
51     loop on ONE box, behind an **edge auth gate** (engine stays `127.0.0.1`, CORS `` is
moot). It is the single-box
52     profile of the engine's decoupled `StorageBackend` + `backend.worker` architecture.
MVP runs on the CPU droplet via
53     **Gemini/stub** ; **real local-GPU detection needs a GPU host** (DO droplet / GCE VM /
Cloud Run+GPU ? owner priority
54     ASAP, runs parallel to the UI MVP). Per-rally selection = one screen. Owner decisions
pending in §10 (edge auth,
55     retention, cost ceiling, GPU host, P5 bridge).
56     - **Alpha SPA** in `web/` (Vite + React + TS + Tailwind) ? hosts the operator console +
the per-rally selection UI above.
57     - **BYO-storage + autosync** ? connect Drive/OneDrive/GCS/S3, watch the folder, auto-
prepare reels
58     (engine `PLATFORM_ARCHITECTURE.md §3`). The form's GPU/storage answers are the demand
signal.
59     - **Control plane** (`control-plane/`) at beta.
60
61     ## Org & repos
62     - ? **DONE (2026-06-20): proprietary repos consolidated into the `Khelsutra` org
(private).** Transferred
63     `badminton-highlight-indexer` (engine), `sports-data-collector`, `sports-obsreport` ?
`Khelsutra/` (stay private; no
64     open PRs/forks; GitHub redirects git ops so active sessions kept working; local
remotes repointed + verified).
65     `rally-annotator` stays in the org **PUBLIC** (OSS MIT plugin the site links to ? do
not make private). Owner is org
66     admin (full access) and can **invite a contributor** (org member / per-repo outside
collaborator) anytime.
67     - ? **`khelsutra` (this site repo) ? owner step pending.** A transfer breaks the
**Cloudflare Workers-Builds** git
68     connection (auto-deploy) until the Cloudflare GitHub App is installed on the
`Khelsutra` org and the Worker
69     reconnected (dashboard). When ready: install the CF app on the org ? `gh api
repos/avidullu/khelsutra/transfer -f
70     new_owner=Khelsutra` ? reconnect the Worker to `Khelsutra/khelsutra` ? verify a
deploy.
71     - ? **Engine CI under the org** ? verify the engine's GitHub Actions run on the next PR
(free orgs default Actions-on;
72     enable in org settings if a run doesn't start).
73
74     ## Convention
75     Substantial work gets a tracked doc in `docs/`; one PR per item off `origin/master`;
merge to `master`
76     auto-deploys (CI must be green).
77

```

15. assistant (2026-06-21T13:56:41.135Z)

I now have all the docs read. Let me verify the current state of the droplet/option-3 parking decision and check whether there's a current-state memory file that confirms the owner just parked it. Let me also confirm the latest commits reflect what the docs claim.

16. user (2026-06-21T13:56:42.417Z)

```

7750afc docs(dry-run): Path-B live e2e DONE ? in-UI bucket?reel CUJ + screencast (DoD §6 box 3
?) (#57)
4525c5d feat(web): B-P2 ? in-UI bucket ingest (point at a gs:// video ? process ? load) (#56)
d0461fd docs: correct stale "bucket ingest is net-new" claims (B-P1 already shipped in M2/#280)
(#55)
2421557 docs(dry-run): record Path-A local result (en+hi reel downloaded) + the engine fix it
surfaced (#54)
113587f docs: set the dry-run CUJ as a tracked goal (UI drives a bucket video -> reel download,

```


multilingual) (#53)
1b69e1a feat(site): show a localized UI screenshot on each academy flyer (#52)
3a42847 feat(web): localized sample-query prompt (en/kn/hi) + i18n-test rule for query parsing (#51)
0021c4f test(web): guard the SPA stays self-contained / offline-renderable (#50)

17. assistant (2026-06-21T13:56:50.440Z)

All confirmed against git history. I have everything needed to produce the structured output. Let me compile the findings for the appliance/CUJ track.

Key facts established:

- **LOCAL_APPLIANCE_ARCHITECTURE.md**: §9 tracker: T0 (?), T1-T11 (?), T12 (? engine done/SPA pending), T13 (? owner ? GPU host), T14 (? owner ? reserved bridge). §10 has 5 blocking owner gates.
- **DRY_RUN_CUJ.md**: §6 DoD: boxes 1-3 (Path A, Hindi, Path B/B-P2) all DONE; box 4 (option 3, CPU droplet) remains, droplet un-provisioned. Owner parked option 3 (GPU machine needs a subscription).
- **PER_RALLY_SELECTION.md**: §7: P0-P5 ? (M0/MVP complete); P6-P7 ? actionable (engine); P8-P10 ? owner-gated.
- **REQUEST_ACCESS_FORM.md**: §6: P0-P1 ?; P2 ? (owner deploy); P3-P4 ?.
- **PORTABILITY.md**: design + proven, no open tracker rows.
- **NEXT_STEPS.md**: active focus is Friendly & Multilingual (separate track); appliance/CUJ next-steps bullets captured.

18. user (2026-06-21T13:57:53.732Z)

Structured output provided successfully

19. assistant (2026-06-21T13:58:05.796Z)

I've extracted the appliance/CUJ track status and next steps from all six docs. Summary of what I found:

Status: The dry-run CUJ goal is largely DONE ? DoD boxes 1-3 (Path A local en+hi, Path B/B-P2 in-UI gs://bucket?reel) all shipped 2026-06-21 with screencasts, and both engine bugs the runs surfaced are fixed (#281, #289). Only box 4 (CPU-droplet confirmation) remains and the owner just PARKED it (droplet un-provisioned, no GPU; a real GPU machine needs a subscription). The SPA back-half it rides on (PER_RALLY_SELECTION M0/MVP, P0-P5) is complete. The broader KhelSutra Studio appliance (LOCAL_APPLIANCE_ARCHITECTURE) is still DRAFT ? only T0 and T12 are in-progress; T1-T11 not started; T13 (GPU host) and T14 (bucket-worker bridge) are owner-gated.

Captured 24 next-step items across the T0-T14 tracker, PER_RALLY_SELECTION P6-P10, REQUEST_ACCESS_FORM P2-P4, and NEXT_STEPS bullets ? each tagged actionable / owner-gated / blocked with effort.

5 owner gates (§10 Q1-Q5: edge-auth, retention, cost ceiling, GPU host, P5 bridge) plus the parked droplet decision.

5 stale/inconsistent flags worth noting (not fixed, just flagged):

- T12 still shows ? but B-P2 SPA ingest shipped (web #56).
- REQUEST_ACCESS_FORM P2 marked ? "owner deploy pending" but NEXT_STEPS says the form is already LIVE on Cloudflare with D1 + email.
- `/api/health` is reconciled as EXISTING in §3 but still listed under "net-new" in §4.6/§11.
- The umbrella doc's DRAFT framing understates shipped slices.
- Header "Last updated: 2026-06-20" dates are stale vs 2026-06-21 body content.

Note: per NEXT_STEPS, the appliance is NOT the current active focus ? the owner's active directive is the separate "Friendly & Multilingual" track (FRIENDLY_AND_MULTILINGUAL.md).